Explanation Shift under Domain Shift

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Abstract

Model explanations are increasingly used to support transparency in high-risk systems, yet their behaviour under distribution shift remains poorly understood. In particular, it remains unclear whether changes in explanations can be reduced to changes in model behaviour, or whether they provide additional information beyond predictive change. This paper examines post-hoc explanations under controlled covariate shift and analyses the relation between predictive change, explanation drift, and explanation-quality metrics. To disentangle these effects, it combines a paired clean-corrupted design with separate analyses at the levels of predictions, explanations, and evaluation scores. Empirically, the study uses CIFAR-10-C, a ResNet-18 classifier, and two structurally different attribution methods, GradCAM and Integrated Gradients, across multiple corruption types and severity levels. The results show that explanation drift increases systematically with corruption severity, but that its magnitude and corruption profile differ qualitatively across attribution methods. Crucially, substantial drift remains even when the analysis is restricted to samples whose predictions are stable and correct, showing that explanation change is not reducible to predictive failure alone. Changes in explanation-quality metrics are likewise not uniform across methods: in the present setup, Sparseness primarily reflects model-behaviour change for GradCAM, whereas for Integrated Gradients it continues to capture attribution-level reorganisation under stable predictions. Taken together, these findings show that performance shift, explanation shift, and metric shift are related but not interchangeable, and that their relation depends on the attribution method through which explanations are observed. Conclusions drawn from a single explainer must therefore be interpreted in light of its specific sensitivity profile.

1 Introduction

Artificial intelligence is increasingly deployed in safety-critical domains such as autonomous driving, medical diagnosis, and critical infrastructure, where even minor system failures can have serious real-world consequences. At the same time, the complexity of such applications is often reduced to simplifying assumptions, most notably that training and deployment data come from the same, or a sufficiently similar, distribution.

Autonomous driving illustrates the fragility of this assumption. Systems trained predominantly on daytime traffic scenes may perform reliably under those conditions, yet struggle at night, under reduced visibility, or in less typical traffic situations. More generally, when the conditions encountered at deployment differ from those seen during training - a phenomenon known as distribution shift - models may no longer generalise reliably, leading to degraded performance, less stable predictions, and unexpected failures. This problem is not confined to autonomous driving: in medical imaging, models trained at one clinical site are often applied under different acquisition conditions, while in critical infrastructure, the statistical properties of monitored systems may drift over time. Distribution shift is therefore not an edge case but a routine challenge of real-world deployment (Koh et al., 2021), and its consequences can extend far beyond degraded accuracy.

As AI systems become more common in such high-stakes environments, the demand for transparency grows. In safety-critical settings, systems must be not only reliable, but also accountable. This development is reinforced by regulatory frameworks such as the EU AI Act, which places particular emphasis on high-risk systems and the need to make decisions and their associated risks transparent (European Union, 2024). This broader push for transparency has been a major driver of explainable artificial intelligence (XAI), which seeks to make model decisions more understandable (Guidotti et al., 2018; Arrieta et al., 2020).

However, if distribution shift alters not only model predictions but also model explanations, the problem cannot be reduced to predictive degradation alone. The central question is therefore whether explanation change simply tracks predictive change or whether it provides additional information under shifting deployment conditions. To address this question, the report first develops a conceptual framework for relating distribution shift, explanation shift, and explanation evaluation (Sections 3.1–3.3), and then studies how predictions, uncertainty, and post-hoc explanations change under controlled shift conditions (Sections 4 and 5). In doing so, it tests whether explanation shift is empirically detectable, whether it is reducible to predictive change, and whether evaluation metrics capture explanation-level change under shift.

2 Background

2.1 Domain Shift

Domain shift describes a mismatch between the data distribution during training and the distribution encountered at deployment. Formally, $P_s(X, Y) \neq P_t(X, Y)$, where P_s denotes the source-domain distribution and P_t the target-domain distribution (Quiñonero-Candela et al., 2009; Moreno-Torres et al., 2012). As long as deployment inputs remain sufficiently similar to the conditions represented during training, they can be treated as *in-distribution*; once this no longer holds, they are considered *out-of-distribution*. Unlike ordinary generalisation, where the challenge is to handle unseen samples from the same distribution, domain shift requires coping with a change in the underlying data distribution itself.

Domain shift does not take a single form. Different types of shift must be distinguished because they affect the data-generating process in distinct ways. In the following, three commonly distinguished forms of shift are introduced (Moreno-Torres et al., 2012).

Covariate shift. Covariate shift occurs when the marginal input distribution changes while the conditional label distribution remains constant:

$$P_s(X) \neq P_t(X), \qquad P_s(Y \mid X) = P_t(Y \mid X).$$

The inputs change in appearance, but the underlying mapping from inputs to labels remains the same. A canonical example is CIFAR-10-C, where standard CIFAR-10 images are subjected to controlled corruptions such as noise, blur, or brightness changes without changing the object categories or their labels.

Label shift. Label shift occurs when the marginal label distribution changes while the class-conditional input distribution remains approximately stable:

$$P_s(Y) \neq P_t(Y), \qquad P_s(X \mid Y) = P_t(X \mid Y).$$

Here, the classes themselves remain similar in their observable characteristics, but their relative frequency changes. In medical applications, for instance, a diagnostic model trained on a general population may be deployed during an epidemic in which the prevalence of a target condition is substantially elevated: the clinical presentation of cases remains similar, but the class balance has shifted.

Concept shift. Concept shift occurs when the conditional label distribution itself changes:

$$P_s(Y \mid X) \neq P_t(Y \mid X).$$

In this case, the mapping from inputs to labels has changed, not merely the distribution of inputs or labels. The same input may therefore be assigned a different label under P_t than under P_s. This can arise when the meaning or definition of the target variable changes, for instance through revised clinical diagnostic criteria or changed regulatory rules (Moreno-Torres et al., 2012).

These types are not mutually exclusive and often co-occur in practice. The distinction nevertheless matters for interpretation, since different forms of shift violate different modelling assumptions and therefore imply different consequences for model behaviour and explanations.

For the present study, the focus is on covariate shift. This form of shift can be introduced in a controlled and gradual manner, is well suited to vision experiments, and allows changes in predictions and explanations to be analysed while the underlying label semantics remain fixed.

2.2 Explainable Artificial Intelligence

Explainable artificial intelligence (XAI) aims to make model decisions more interpretable (Guidotti et al., 2018; Arrieta et al., 2020). Its relevance becomes particularly clear under domain shift, where predictive performance alone does not reveal whether a model continues to rely on the same evidence as before. A model may therefore still produce correct predictions even though the basis for these predictions has changed. Explanations are useful precisely because they make this shift in decision basis visible by indicating which parts of the input the model relies on.

XAI does not denote a single explanatory technique, but a heterogeneous set of approaches that differ in both form and purpose (Guidotti et al., 2018; Arrieta et al., 2020). Broadly speaking, these approaches vary along three axes. First, interpretability may be built directly into the model or applied post hoc to explain decisions after training. Second, explanations differ in scope, ranging from local explanations of individual predictions to global characterisations of overall model behaviour. Third, they differ in what they treat as explanatory evidence, relying for example on input features, similar training instances, or higher-level human-interpretable concepts. Because these approaches address different questions, they are suited to different analytical purposes.

The present work focuses on post-hoc, local, feature-based explanations. Generated after training for a specific prediction, such explanations indicate which input features or regions are treated as relevant for that decision. This makes them well suited to the vision setting considered here, where highlighted evidence can be inspected directly and compared across inputs.

2.3 XAI Evaluation

Unlike predictive performance, explanation quality usually cannot be assessed against directly observable ground truth. In most cases, there is no uniquely correct attribution, and qualitative inspection alone is not sufficient, since visually plausible explanations may still fail to reflect the model's actual behaviour (**?**Guidotti et al., 2018; Arrieta et al., 2020; Adebayo et al., 2018; Dombrowski et al., 2019).

XAI evaluation therefore proceeds indirectly. Rather than verifying a single "correct" explanation, it asks whether an explanation satisfies properties that are commonly associated with explanation quality and turns these properties into measurable scores (Nauta et al., 2023; Hedström et al., 2023). Following Bommer et al. (2024), we organise the evaluation around four such properties. *Faithfulness* asks whether the explanation reflects the factors that actually drive the model's prediction. *Robustness* captures whether the explanation remains stable under small, non-substantive perturbations of the input. *Complexity* addresses whether the attribution is compact and interpretable rather than diffuse or noisy. *Randomisation* tests whether the explanation depends on meaningful model behaviour by checking whether it remains sensitive to the destruction of learned model or label structure. These properties capture distinct aspects of explanation quality and are therefore complementary rather than interchangeable (Bommer et al., 2024; Nauta et al., 2023). Further properties such as localisation and sanity are also established in the literature, but are not the focus here.

Because these properties are not directly observable, they must be operationalised through proxy tasks and evaluation protocols (Nauta et al., 2023; Hedström et al., 2023). In practice, this is done by applying a controlled intervention to the input, the model, or the explanation and measuring how the prediction or attribution responds. XAI metrics therefore do not measure explanation quality directly, but quantify it relative to a particular evaluation setup. Different metrics target different properties, and even metrics aimed at the same property may behave differently depending on how that property is operationalised (Nauta et al., 2023; Hedström et al., 2024). Explanation quality can therefore not be reduced to a single universal score. In practice, multiple metrics are often needed to obtain a more complete picture of explanation behaviour.

To support reproducible evaluation under these conditions, Quantus provides a standardised framework for implementing a broad range of XAI metrics under a common interface, thereby improving comparability across studies (Hedström et al., 2023). A related question is how reliable such metric scores are themselves. MetaQuantus addresses this through a meta-evaluation framework that assesses metrics along two complementary dimensions: *noise resilience*, the extent to which scores remain stable under minor perturbations that do not change the prediction, and *adversary reactivity*, the extent to which scores respond appropriately to disruptive perturbations that do (Hedström et al., 2024). The trustworthiness of metric scores is therefore itself an empirical question.

Important limitations nevertheless remain: since explanations generally lack direct ground truth, the validity of explanation evaluation cannot itself be established straightforwardly, and metric scores remain sensitive to the design of the evaluation protocol. These limitations become especially important once explanations are evaluated under changing input conditions, as in the setting considered here.

3 Conceptual and Methodological Framework

To understand the effects of distribution shift on post-hoc explanations, the relevant components of the explanation pipeline must be considered jointly under shifted deployment conditions. This requires a holistic view of the interdependent stages through which model behaviour is transformed into predictions, explanations, evaluation outcomes, and meta-evaluation, rather than treating these components in isolation.

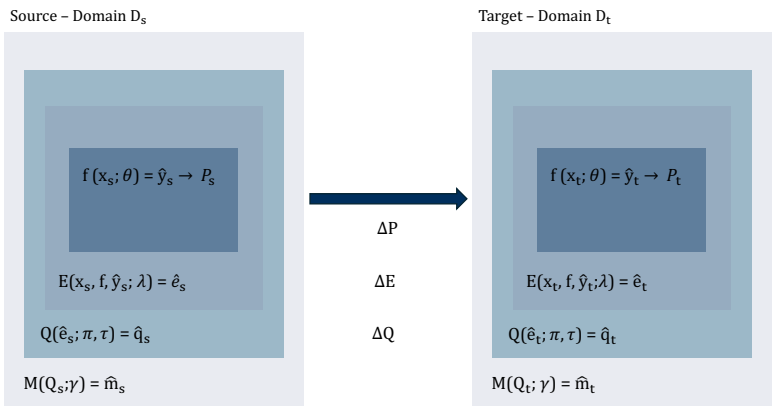

Figure 1: Stage-wise dependency structure across source and target domains, giving rise to ΔP, ΔE, and ΔQ.

3.1 Pipeline Under Domain Shift

Figure 1 summarizes these dependencies between the individual components as a multi-stage pipeline, with information flowing sequentially through prediction, explanation, evaluation, and meta-evaluation. It builds on the conditional dependency structure formalized by Hedström et al. (2024). The framework used here adapts the notation and extends this structure to the domain-shift setting by explicitly distinguishing between a source domain D_s and a target domain D_t. This framework helps to make explicit the stages at which changes may arise under distribution shift.

Let D denote a deployment domain, and let f denote a trained predictive model with parameters θ. For an input x drawn from D, the model produces a prediction

$$\hat{y} = f(x; \theta).$$

We define $P(D)$ as the task-level performance (e.g. accuracy) of f on D. Performance can be directly assessed against ground-truth labels and is therefore directly verifiable.

To explain a prediction, a post-hoc attribution method E produces a local explanation $\hat{e} = E(x, f, \hat{y}; \lambda)$. The result, such as a saliency map, describes the evidence the method identifies as relevant for that prediction. The explanation is jointly determined by the input x, the model f, and the prediction $\hat{y}$. Unlike predictions, explanations generally lack ground-truth labels and thus occupy what Hedström et al. (2024) call the *unverifiable space*.

The resulting explanation is then assessed indirectly through an evaluation metric, written as $\hat{q} = Q(\hat{e}; \pi, \tau)$, where π denotes the evaluation protocol and τ evaluator-specific parameters. Because the quality estimator is applied to the output of an unverifiable explanation function, the evaluation outcome inherits this unverifiability. Hedström et al. (2024) refer to this dependence of the evaluation stage on the explanation stage as the *Challenge of Unverifiability*. At the same time, $\hat{q}$ is not a direct observation of explanation quality, but an operationalised quantity defined relative to a specific protocol and domain setting.

At the end of this chain stands meta-evaluation M, written as $\hat{m} = M(Q; \gamma)$, where γ denotes meta-evaluation parameters. Following Hedström et al. (2024), meta-evaluation is understood here as the evaluation of the evaluation method itself. Since the validity of Q cannot be established directly in the absence of ground-truth explanation labels, this stage is concerned with the reliability of Q as an estimator under controlled conditions. In the present framework, M therefore characterises how reliably the evaluation method behaves under controlled perturbation-based tests.

The stage-wise dependency structure also makes explicit that the individual components rest on distinct assumptions: learned decision boundaries in f, attribution logic in E, distributional requirements in Q, and meta-evaluation criteria in M. Because these stages are sequentially dependent, uncertainty can propagate through the chain, and distortions at one level may affect subsequent stages.

To analyse this structure under distribution shift, we distinguish between a source domain D_s and a target domain D_t. This yields three quantities that capture change at different levels of the pipeline: ΔP, ΔE, and ΔQ. While ΔP and ΔQ can be defined as differences between aggregate scalar scores, ΔE must be operationalised through a divergence or similarity measure over attributions.

3.2 Expected Relations Between ΔP and ΔE

Given the dependency structure introduced in Section 3.1, the question arises how performance and explanations relate under shift. An intuitive expectation is that the two should broadly move together, with ΔE increasing or decreasing alongside ΔP: the smaller the performance change, the smaller the explanatory change, and vice versa. In that case, ΔE would not constitute an independent quantity, but would largely be reducible to ΔP as another expression of the same underlying distributional change.

This proportional view, however, is stronger than what the pipeline structure warrants. As established in Section 3.1, an explanation depends not only on the model and the prediction, but also on the input under consideration. Under covariate shift, changes in the input across domains should therefore induce some change in the explanation, even if the model f remains fixed. A more defensible expectation is semantic coherence rather than strict proportionality: if the model reaches the same prediction for the same functionally relevant reasons, the explanation should continue to highlight the evidence underlying that prediction rather than drift toward unrelated or spurious structure.

This weaker expectation already implies that ΔP and ΔE capture different aspects of model behaviour. ΔP describes change at the level of predictive performance, whereas ΔE describes change at the level of the attributed decision basis. Although both are induced by the same domain shift, they are not interchangeable. This motivates the notion of *explanation shift*. Following Mougan et al. (2025), we use this term for the general phenomenon that explanations change under distribution shift. In the empirical analysis, we quantify this phenomenon as *explanation drift*, denoted by ΔE.

Prior work confirms that the coupling between ΔP and ΔE does not hold in general. Explanations have been shown to drift under shift even when predictions remain correct or aggregate performance appears relatively stable (Mougan et al., 2025; Cossu et al., 2024). We refer to this divergence as *decoupling*.

3.3 Mechanisms of Decoupling

Decoupling between performance, explanations, and evaluation scores can arise at different points in the pipeline. For the empirical analyses in Section 5, two broad sources of divergence are distinguished: measurement and comparability problems in the evaluation protocol π, and genuine decoupling between model behaviour and explanation behaviour in the relation between f and E. While not an exhaustive taxonomy, these analytical categories help separate superficially similar empirical patterns that may reflect different underlying mechanisms.

Measurement and comparability problems. One source is that the observed divergence may not reflect a property of the model or the explanation, but an artefact of the evaluation protocol. Perturbation-based metrics, for instance, rely on operations such as masking, feature removal, or baseline substitution that are designed under implicit assumptions about the input distribution. Under shift, these assumptions may no longer hold: the perturbed samples can become implausible under either distribution, so that the resulting score reflects not only an explanation property but also the model's reaction to an input it was never trained on (Hase et al., 2021; Qiu et al., 2021). Even without perturbation, absolute metric values may cease to be comparable across domains when feature scales, correlations, or marginal statistics change. A difference in evaluation scores between D_s and D_t may then indicate not a change in explanation quality but a change in the distributional conditions under which the score is computed. A natural response is to move from absolute scores to rankings. However, this does not resolve the problem as different metrics within the same quality category, such as faithfulness, can produce conflicting orderings of the same explanation methods even within a single domain (Klein et al., 2024; Hedström et al., 2024). Under shift, where the evaluation context itself changes, such inconsistencies are likely to compound.

Genuine decoupling. Another source lies in an actual decoupling between model behaviour and explanation behaviour. Under shift, the model's weights remain fixed, but the features that are most predictive can change with the input distribution. A model that relied on shape under D_s may effectively weight texture or background statistics more heavily under D_t, not because it was retrained, but because these features have become stronger correlates of the label in the shifted distribution (Geirhos et al., 2020). This change in model strategy may remain invisible in the aggregate performance $P(D_t)$ if overall accuracy is maintained. Whether it becomes visible in the explanation depends on the explanation method. This dependence can give rise to three distinct patterns of decoupling:

- *Explainer inertness:* The model's decision basis has changed, but the explainer responds only weakly. This occurs, for instance, when the attribution method operates on output logits or local gradients that change little even when intermediate representations have shifted. The explanation appears stable, providing a consistent but increasingly inaccurate account of the model's behaviour. This pattern produces a false sense of reassurance.

- *Explanation drift with stable prediction:* The converse pattern occurs when predictions remain correct, but explanations shift substantially because the model achieves the same output through different feature combinations under the shifted distribution (Cossu et al., 2024; Li et al., 2023). In such cases, predictive performance alone does not reveal that the explanation has drifted.

- *Masked degradation:* Unlike the preceding two patterns, predictive performance degrades in this case, while explanations remain comparatively stable. Apparent explanatory stability therefore masks predictive degradation. As with explainer inertness, stability here is not reassuring but actively misleading.

More broadly, Li et al. (2023) show that standard robustness methods can improve predictive performance while making explanations less stable, directly confirming that accuracy robustness and explanation robustness are independent properties.

Consequences for interpretation. An observed divergence between ΔP, ΔE, and ΔQ can therefore not be interpreted uniformly. It may originate from a protocol artefact, a genuine change in model strategy, a mismatch in explainer sensitivity, or from several of these simultaneously. Because

distribution shift acts on a multi-stage pipeline in which each level can respond differently, neither predictive performance nor explanation metric scores can by themselves determine whether the model's decision basis, the explanation, and its evaluation remain aligned. In addition, any conclusion drawn from such patterns must remain limited to the specific model, explanation method, and evaluation protocol under which they were obtained. What is required instead is a layered interpretation that considers each level of the pipeline and its potential failure modes. The experimental analysis that follows is designed to make such an interpretation possible.

4 Experimental Design and Evaluation

4.1 Aim of the Experiment

This experiment characterises how post-hoc explanations behave under controlled covariate shift and how this behaviour relates to predictive change and to the evaluation metrics defined over attributions. To this end, it separates explanation drift from accompanying changes in model predictions, allowing the effects of shift strength, model behaviour, and attribution method to be examined independently.

The analysis is structured around three research questions: (1) Does explanation drift arise systematically under increasing corruption severity, and which factors determine its magnitude and pattern across attribution methods? (2) Can explanation drift be reduced to changes in predictive behaviour, or does it constitute an independent signal? (3) Do changes in explanation-quality metrics under shift reflect changes in the explanations themselves, or are they instead driven primarily by changes in model behaviour and by properties of the evaluation protocol?

Taken together, these questions establish whether explanation drift under distribution shift is systematic, behaviourally informative beyond predictive accuracy, and consistent across attribution methods or instead method-specific.

4.2 Experimental Setup

Dataset. The study uses CIFAR-10 (Krizhevsky, 2009) as the in-distribution reference and CIFAR-10-C (Hendrycks and Dietterich, 2019) to introduce controlled covariate shift. CIFAR-10-C applies common corruptions to each CIFAR-10 test image at multiple severity levels, producing paired clean-corrupted samples suitable for direct comparison of attributions. The analysis focuses on four corruption types Gaussian noise, fog, brightness, and defocus blur at severity levels 1, 2, 3, and 5 to cover mild through pronounced shift conditions. Figure 2 illustrates this paired setup for one example image under fog corruption.

Figure 2: Example of a paired clean-corrupted sample from CIFAR-10-C under fog corruption. Images are upscaled for visibility.

Model. The predictive model is a ResNet-18 classifier (He et al., 2016) trained on CIFAR-10, implemented via the standard `torchvision` architecture. ResNet-18 was chosen because it offers a suitable balance between representational capacity and computational efficiency. As a robustness check, all experiments were repeated with five random seeds (7, 42, 52, 128, 1200), while keeping all other training settings fixed.

Explanation methods. Two post-hoc explanation methods are used: Gradient-weighted Class Activation Mapping (GradCAM) and Integrated Gradients (IG). *GradCAM* (Selvaraju et al., 2017) produces region-level attributions by weighting convolutional feature maps with gradients with respect to the target class, highlighting the spatial regions that contribute most strongly to the prediction. *Integrated Gradients* (Sundararajan et al., 2017), by contrast, produces pixel-level attributions by accumulating gradients along a path from a baseline to the input. Using two conceptually different explanation methods makes it possible to distinguish method-specific artefacts from more systematic drift patterns.

Pipeline and measured quantities. A modular pipeline executes all combinations of seed, explanation method, corruption type, and severity level. Intermediate artefacts such as predictions, logits, attribution maps, and metric scores are stored at each stage. Measurements are taken at four levels corresponding to the pipeline from Section 3.1: shift strength (MMD^2), predictive behaviour and uncertainty, explanation similarity between paired clean-corrupted attributions, and explanation quality scores computed using Quantus, with additional metric diagnostics based on MetaQuantus (Hedström et al., 2023). How these quantities are operationalized as ΔP, ΔE, and ΔQ is described in Section 4.4. A compact overview of all computed quantities, their interpretation, and their level of aggregation is provided in Appendix A.1.

4.3 Experimental Design

Paired comparison. Since each corrupted image is directly linked to its clean counterpart, predictions and explanations can be compared on a sample-by-sample basis. The paired structure makes it possible to analyse what changes for the same sample under corruption, rather than inferring change only from differences between aggregate distributions.

Corruption-wise and severity-wise analysis. Results are analysed separately by corruption type and severity level. The corruption-wise view reveals whether explanation drift follows different patterns across forms of degradation, while the severity-wise view shows whether such effects scale gradually with shift strength or emerge only beyond a threshold. MMD^2 is included as an explicit measure of distributional distance, so that comparisons are anchored in measured shift magnitude rather than nominal severity labels alone.

Slice-based analysis. To separate explanation drift from prediction failure, the analysis uses three nested subsets. *All samples* provides the aggregate view. *Invariant predictions* isolates pairs where the predicted class does not change under corruption, so that any observed explanation drift occurs despite an unchanged output. *Both-correct predictions* further restricts to pairs where the model classifies both inputs correctly, the strictest condition, under which explanation drift cannot be attributed to prediction error on either input.

4.4 Evaluation Procedure

The evaluation operationalizes the pipeline introduced in Section 3.1 at four levels: shift strength, predictive behaviour, explanation drift, and explanation-related metric scores. All quantities are computed per seed and then aggregated, reporting means and variability. A detailed overview of the computed quantities is provided in Appendix A.1.

Shift strength. Distributional distance between clean and corrupted representations is quantified using MMD^2.

ΔP: predictive change. Predictive behaviour is analysed at the sample level on each of the three slices defined in Section 4.3. At the aggregate level, predictive change is computed as $\Delta P = P(D_t) - P(D_s)$, the difference in accuracy across domains. Additional uncertainty-related quantities are recorded as complementary indicators.

ΔE: explanation drift. Explanation drift is computed for each paired sample by comparing clean and corrupted attribution maps using three similarity measures: Spearman rank correlation, cosine similarity, and top-k IoU, which capture different aspects of attribution agreement.

ΔQ: explanation-quality score change. Explanation-quality scores are computed using Quantus (Hedström et al., 2023), covering faithfulness, sparseness, robustness and randomisation. For a given metric m, aggregate score change is computed as $\Delta Q_m = \bar{q}_{m,t} - \bar{q}_{m,s}$ for each corruption and severity level.

4.5 Reported Analyses and Scope Restrictions

Not all computed measures are reported in the main analysis. Numerically unstable, computationally infeasible, or highly redundant analyses were either moved to the appendix or omitted entirely.

Shift axis and corruption set. MMD^2 is not reported in the main analysis, since it proved impractical at the required granularity and did not add interpretive value beyond corruption-wise and severity-wise comparisons. Among the four analysed corruption types, defocus blur is omitted from the main results because its behaviour closely resembled fog.

Primary measures for ΔP and ΔE. The predictive-change and explanation-drift quantities introduced above were highly redundant within each group. The main analysis therefore reports one primary measure for ΔP and one for ΔE. The specific metric choices are introduced in Section 5.

ΔQ subset. The reported ΔQ analysis is restricted to the subset of metrics that remained stable and feasible under the full experimental setup. Sparseness is used as the primary metric in the main analysis, while Faithfulness Correlation is reported only in the appendix; further technical details are given in Appendix B.

Slice selection. At the slice level, the *invariant* and *both-correct* subsets produce near-identical drift patterns. The reported analysis therefore uses the stricter both-correct slice, since it excludes both prediction changes and prediction errors.

5 Results and Discussion

This section reports the empirical results for the three research questions introduced in Section 4.1. It first establishes the primary measures for explanation drift and confidence change used throughout the main text.

Metric choices. These choices are based on the correlation analysis reported in Appendix C (Figure 13). The same overall pattern appears both in the pooled analysis and when correlations are broken down by corruption type.

For explanation drift, the three similarity-based measures are largely interchangeable. The only consistent exception is GradCAM, for which $1 - \text{IoU}$ is only moderately correlated with the two continuous similarity measures. This suggests that IoU captures an aspect of map change that the other two measures smooth over. A plausible explanation is structural: GradCAM maps are comparatively coarse and spatially diffuse, so overlap in the most salient regions can change even when whole-map similarity remains fairly stable, whereas IG attributions are higher-resolution and sparser, making the three measures more similar. We therefore adopt $\Delta E = 1 - \rho$ as the primary measure of explanation drift throughout the main text and retain $1 - \text{IoU}$ as a robustness check for GradCAM in Appendix C.

The same redundancy pattern holds for the three confidence-drift measures, which correlate at 0.89–0.99 across explainers, corruptions, and slices. We therefore use $|\Delta H|$, the absolute change in predictive entropy, as the primary confidence-drift measure and do not report the other two separately.

Importantly, explanation drift and confidence drift are only moderately correlated in the pooled analysis. Explanation drift therefore cannot be reduced to changes in model confidence and should be treated as a separate quantity.

All reported values are averages over five independent seeds. Seed variation is small throughout: standard deviations on ΔE stay below 0.03 in every condition and are below 0.015 in most. The seed bands therefore do not alter the qualitative pattern, and we report mean trajectories throughout.

With these measurement choices in place, we first examine whether explanation drift increases systematically under distribution shift and how strongly it scales with corruption severity.

5.1 RQ1: Explanation drift increases with corruption severity

Figure 3 reports $\Delta E = 1 - \rho$ on the full sample set as a function of corruption severity, separately for each explainer and corruption type. For both GradCAM and IG, explanation drift increases monotonically with severity in all reported conditions.

At the same time, the size and shape of the drift curves depend strongly on the corruption type, and this dependence differs across explainers. For GradCAM, the corruption ordering is stable at every severity level: brightness is consistently the mildest corruption ($0.03 \rightarrow 0.35$ from sev $= 1$ to sev $= 5$), whereas Gaussian noise is the most disruptive ($0.38 \rightarrow 0.86$).

IG shows the same monotonic increase with severity, but a different corruption profile. Gaussian noise already induces substantial drift at low severity, reaching $\Delta E \approx 0.51$ at sev $= 1$ and increasing by only $+0.17$ up to sev $= 5$. By contrast, brightness and fog start from much lower levels but increase much more strongly across severity, by $+0.56$ and $+0.54$, respectively, reaching comparable drift levels only at the highest corruption strengths. Gaussian noise is therefore already highly disruptive for IG at low severity, while brightness and fog account for most of the severity-driven increase.

The explainer comparison is therefore not just quantitative but structural. GradCAM is most sensitive to Gaussian noise and comparatively insensitive to brightness, whereas for IG the three

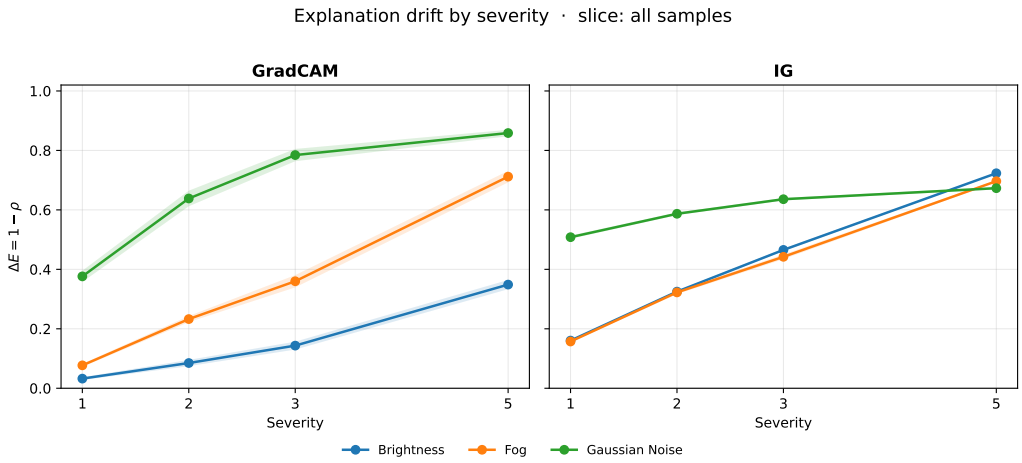

Figure 3: Primary explanation-drift result measured as $\Delta E = 1 - \rho$ on the all-samples slice. Higher values indicate greater drift between clean and corrupted explanations.

corruptions converge to within 0.05 of each other at sev = 5, with brightness and fog accounting for almost all of the severity-driven change.

A plausible explanation is that the two explainers depend on different aspects of the network's processing. GradCAM is based on late-stage feature maps, which appear relatively robust to brightness changes, whereas pixel-level noise disrupts the convolutional stack on which those maps depend. IG, by contrast, integrates gradients along a path from a baseline to the input, so global intensity changes affect that path even when the prediction stays the same, while pixel noise is partly averaged out by the integration.

A second notable pattern is that IG shows higher drift than GradCAM already at sev = 1 in every condition, indicating that even minimal corruption produces a measurable IG response.

Taken together, these results show that explanation drift under distribution shift is systematic and substantial for both attribution methods. However, its shape and magnitude do not follow a single corruption ordering; they depend qualitatively on the explainer. This conclusion is robust to the choice of similarity measure Appendix C.

5.2 RQ2: Explanation drift is only partially coupled to prediction behaviour

Drift across slices. Figure 4 compares explanation drift, $\Delta E = 1 - \rho$, across three nested slices of the sample set: the full set (*all*), the subset with unchanged predictions (*invariant*), and the stricter subset with unchanged and correct predictions (*both-correct*). In seven of the eight explainer $\times$ corruption panels, the three trajectories are nearly identical across all severities; the largest gap between the full set and *both-correct* is only 0.038 (GradCAM, fog, severity 5). Removing samples whose predictions flip under corruption therefore leaves explanation drift almost unchanged. Most of the observed drift is thus not generated by cases in which the corruption changes the prediction.

The main exception is GradCAM under Gaussian noise. Here the full-sample curve lies visibly above the controlled slices at higher severities, with the largest gaps at severities 3 and 5 (0.784 vs. 0.703 and 0.858 vs. 0.781). For GradCAM, Gaussian noise is the strongest corruption at every severity, causing the most prediction flips and therefore the largest reduction when moving to the controlled slices. These flipped samples do contribute additional drift. However, even after they are removed,

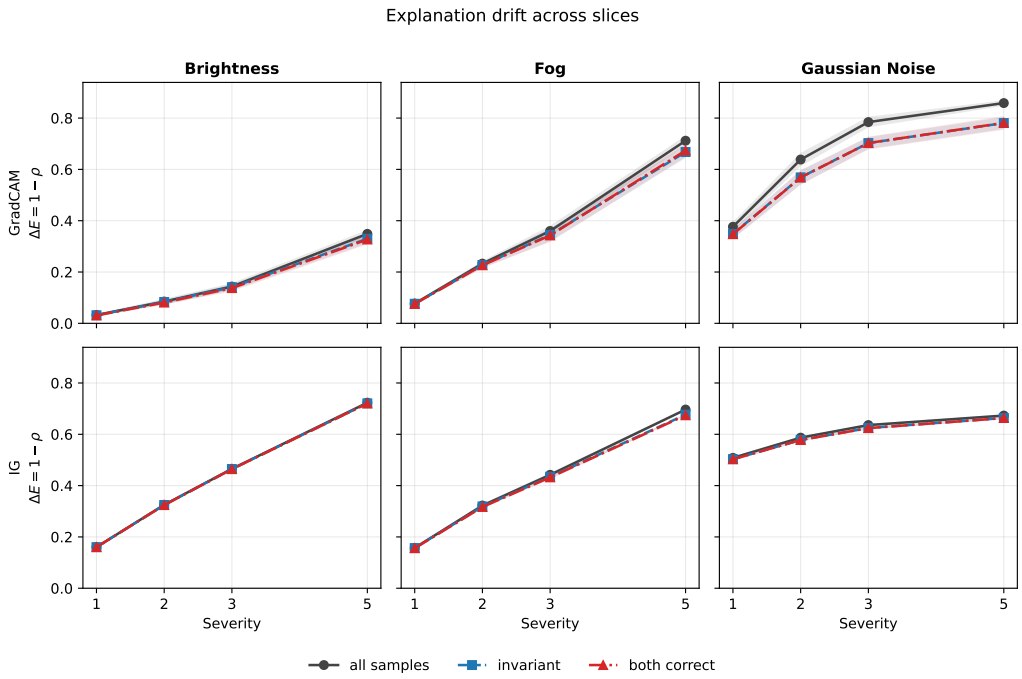

Figure 4: Explanation drift $\Delta E = 1 - \rho$ across the full, invariant, and both-correct slices. The close overlap of the three trajectories shows that most drift remains after prediction-changing samples are removed.

the *both-correct* slice still reaches $\Delta E = 0.781$ at severity 5, near the top of the scale. Prediction changes therefore explain part of the effect in this one configuration, but not most of it. Overall, the slice analysis shows that explanation drift is only partially coupled to prediction behaviour.

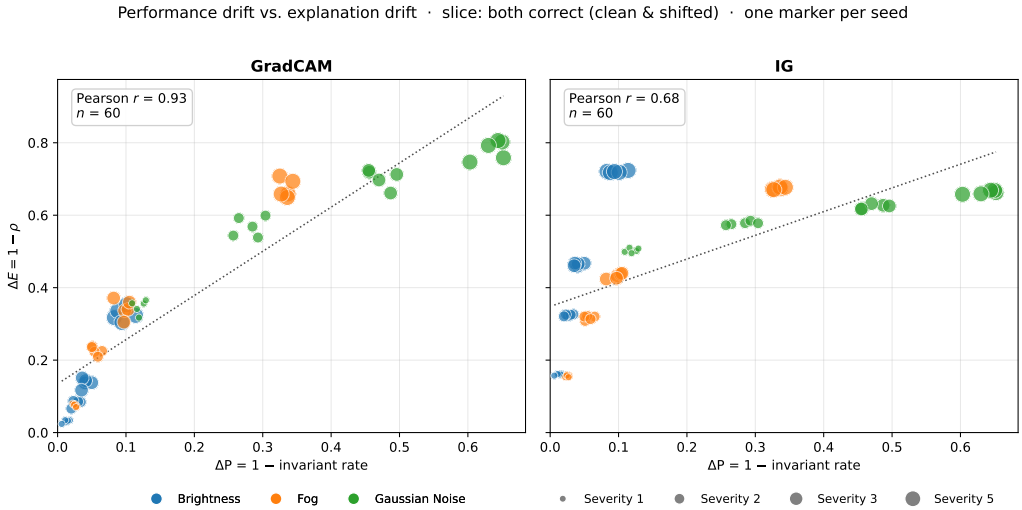

Figure 5: Condition-level relation between predictive change (ΔP; flip rate) and explanation drift ($\Delta E = 1 - \rho$) on the both-correct slice. Each point is one corruption–severity condition.

Across vs. within conditions. Figure 5 and Figure 4 show the same result at two different levels. In Figure 5, each point represents an entire corruptionseverity condition, and ΔP and ΔE are positively associated across those conditions, strongly for GradCAM ($r = 0.93$) and moderately for IG ($r = 0.68$). This between-condition association is therefore not identical across explainers: corruption severity acts as a common driver across conditions, tending to increase both prediction loss and explanation drift, but the strength of that co-movement differs between the two attribution methods. Figure 4, by contrast, shows what happens *within* a fixed condition when prediction-changing samples are removed. There, drift usually remains. The cross-condition association therefore does not imply that explanation drift is caused by prediction change at the sample level.

This separation is especially visible in the IG scatter, which contains a brightness cluster at $\Delta P \approx 0.08$ and $\Delta E \approx 0.72$: brightness leaves predictions largely intact while still inducing near-maximal explanation drift. This is the clearest case in Figure 5 where predictive change remains small but explanation drift is already very large.

Sample-level view: four behavioural regimes. The slice curves average over samples and therefore do not show how drift is distributed across individual cases. Figure 6 addresses this by classifying each paired sample by prediction outcome and drift magnitude. Using a global high-drift threshold at the 75th percentile of the pooled drift distribution ($q_{0.75} = 0.632$), samples fall into four regimes: *Robust* (correct and low drift), *Silent Drift* (correct and high drift), *Visible Failure* (incorrect and high drift), and *Hidden Failure* (incorrect and low drift). All regime shares reported here use the $1 - \rho$ metric; Appendix D reports the corresponding comparison under $1 - \text{IoU}$.

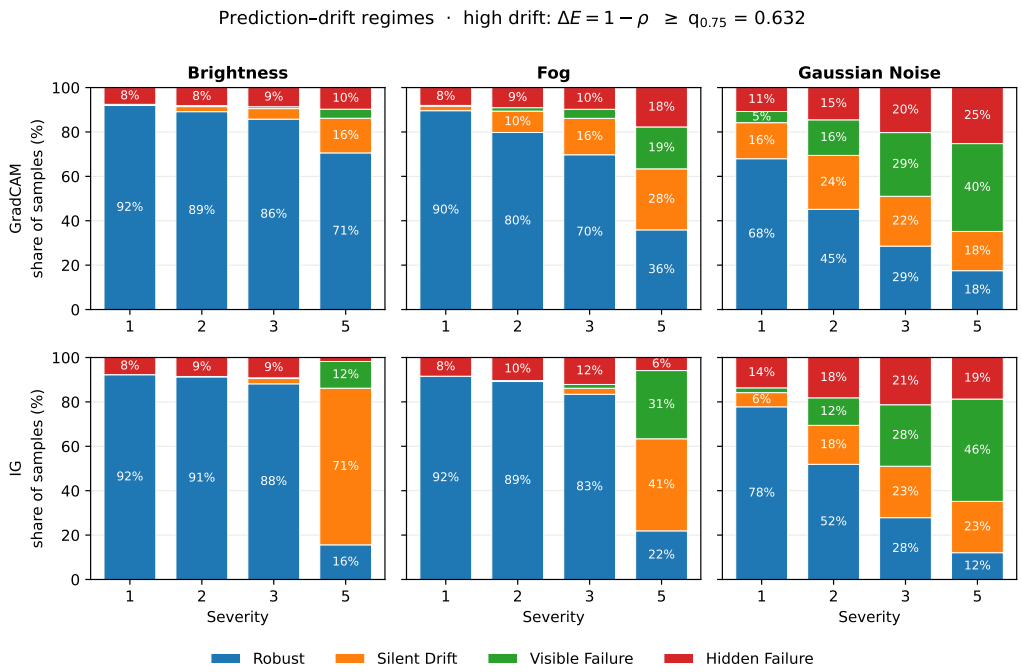

Figure 6: Prediction-drift regime shares across severities under $\Delta E = 1 - \rho$ ($q_{0.75} = 0.632$). Samples are grouped by prediction correctness and drift magnitude into Robust, Silent Drift, Visible Failure, and Hidden Failure.

The key regime for decoupling is *Silent Drift*: the prediction remains correct, but the explanation has already changed substantially. Its share increases with severity in every panel and is especially pronounced for IG under photometric corruptions. At severity 5, IG under brightness places 70.6% of samples in Silent Drift, with only 15.6% remaining Robust; under fog, Silent Drift still reaches 41.5%. For GradCAM, the corresponding shares are smaller but still substantial: 15.6% under brightness and 27.5% under fog. Under Gaussian noise, the pattern shifts because prediction flips become frequent: mass moves out of Robust and Silent Drift into Visible Failure (IG 46.1%; GradCAM 39.6%) and Hidden Failure (IG 18.7%; GradCAM 25.2%), yet Silent Drift still accounts for 23.1% of IG and 17.7% of GradCAM samples. Decoupling is therefore not just an aggregate residual; in several settings, it characterises a substantial share of individual samples.

Because both explainers are applied to the same model on the same samples, the total prediction failure rate is identical within each condition. What differs is whether those failures are accompanied by a strong drift signal. Under brightness at severity 5, both explainers register 13.8% total failures, but they partition those failures very differently: for IG, 87% of failures are accompanied by high drift, whereas for GradCAM 70% of failures remain in the low-drift regime. The same asymmetry appears under fog, though less strongly: the Hidden Failure share among failures is 16% for IG and 48% for GradCAM. Under Gaussian noise, the pattern reverses, with GradCAM becoming the more drift-sensitive explainer. The relevant result is therefore not that one explainer registers more failures than the other, but that the choice of explainer determines which failures are visible to drift-based monitoring.

Distribution shape. Figure 7 plots the distribution of per-sample ρ values for all samples in the *both-correct* slice. The two explainers show qualitatively different shapes. GradCAM remains concentrated near $\rho \approx 1$, while a smaller subset of samples moves far toward low similarity as severity increases. IG, by contrast, shifts downward more evenly across much of the distribution. In other words, GradCAM drift is concentrated in a minority of strongly affected samples, whereas IG drift is spread more broadly across the population. This helps explain why the two explainers can show similar mean drift values while still exhibiting different failure profiles.

Qualitative examples. Figure 8 illustrates the Silent Drift regime with one example per explainer, both drawn from fog at severity 3 on the *both-correct* slice. In both cases, the prediction remains correct before and after corruption, but the attribution changes substantially. For IG, a commercial aircraft shows a clear reorganisation of the saliency map, with per-sample $\rho = 0.339$. For GradCAM, the image of a perched bird exhibits a rarer but much stronger change: the corrupted CAM is almost the spatial inverse of the clean one ($\rho = -0.887$) despite unchanged correct prediction. These examples mirror the distribution-level difference in Figure 7: IG tends to drift more broadly and moderately across samples, whereas GradCAM is stable for many samples but can change abruptly in a smaller subset. Explanation stability therefore cannot be inferred from prediction stability alone.

Pre-shift confidence does not predict later explanation drift. A natural follow-up question is whether drifting samples can be identified in advance from the model's clean-image confidence. Figure 9 tests this by plotting per-sample ΔE against clean maximum softmax probability (Clean MSP) on the *both-correct* slice. Across all severities and both explainers, Pearson correlations remain near zero, in the range $[-0.064, +0.034]$. Clean confidence is therefore not a useful predictor of later explanation drift.

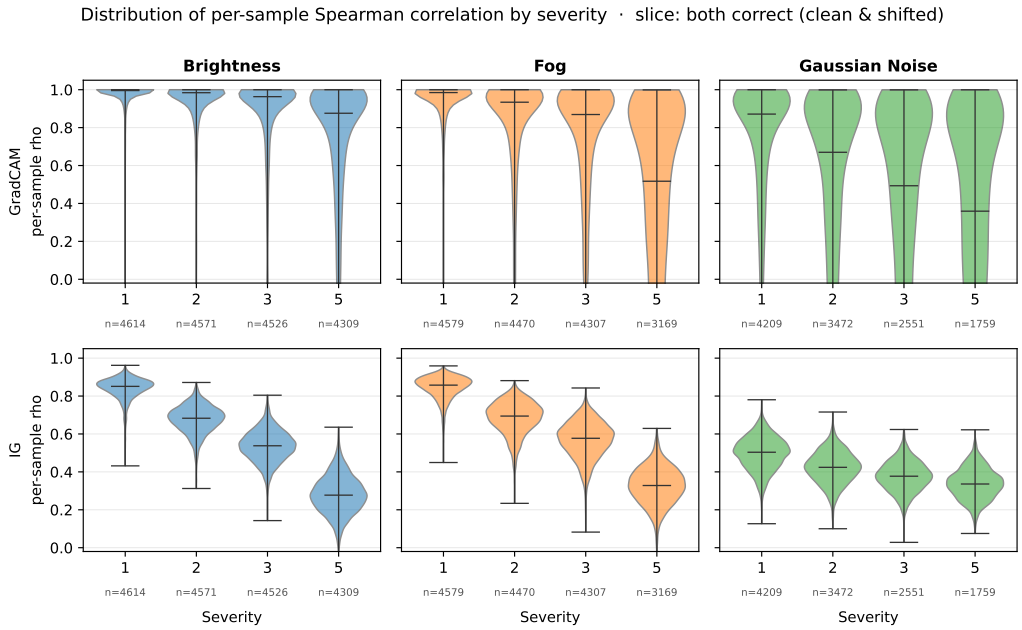

Figure 7: Distribution of per-sample Spearman similarity ρ on the both-correct slice across severities. Lower values indicate stronger attribution drift despite stable correct predictions.

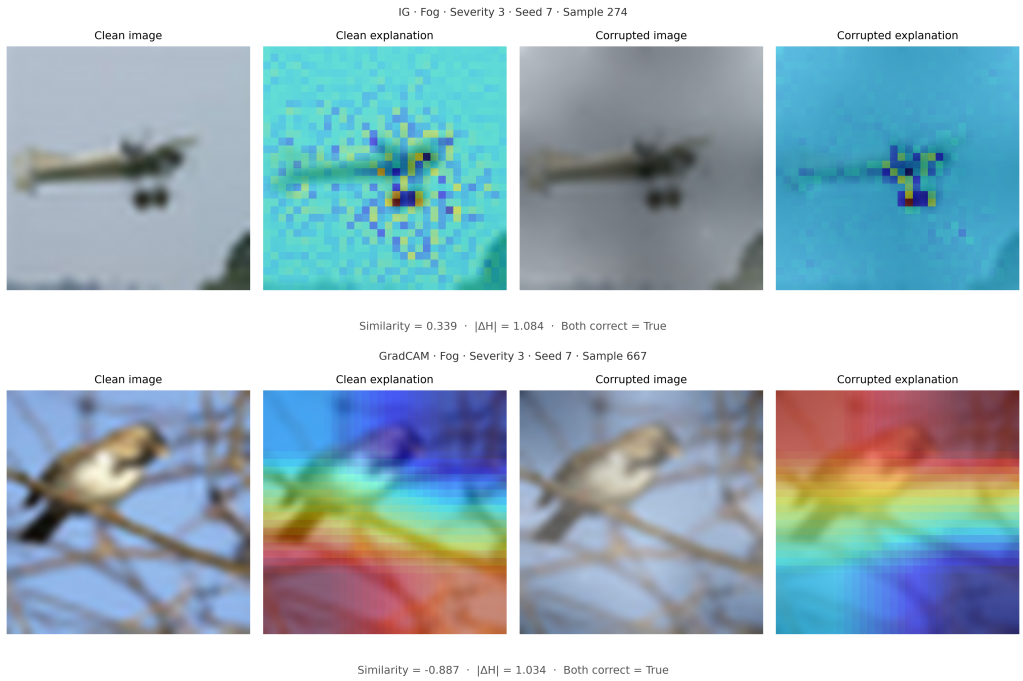

Figure 8: Examples of Silent Drift on the *both-correct* slice under fog at severity 3. The IG example shows substantial attribution reorganisation ($\rho = 0.339$), whereas the GradCAM example shows a near-inverted corrupted CAM ($\rho = -0.887$).

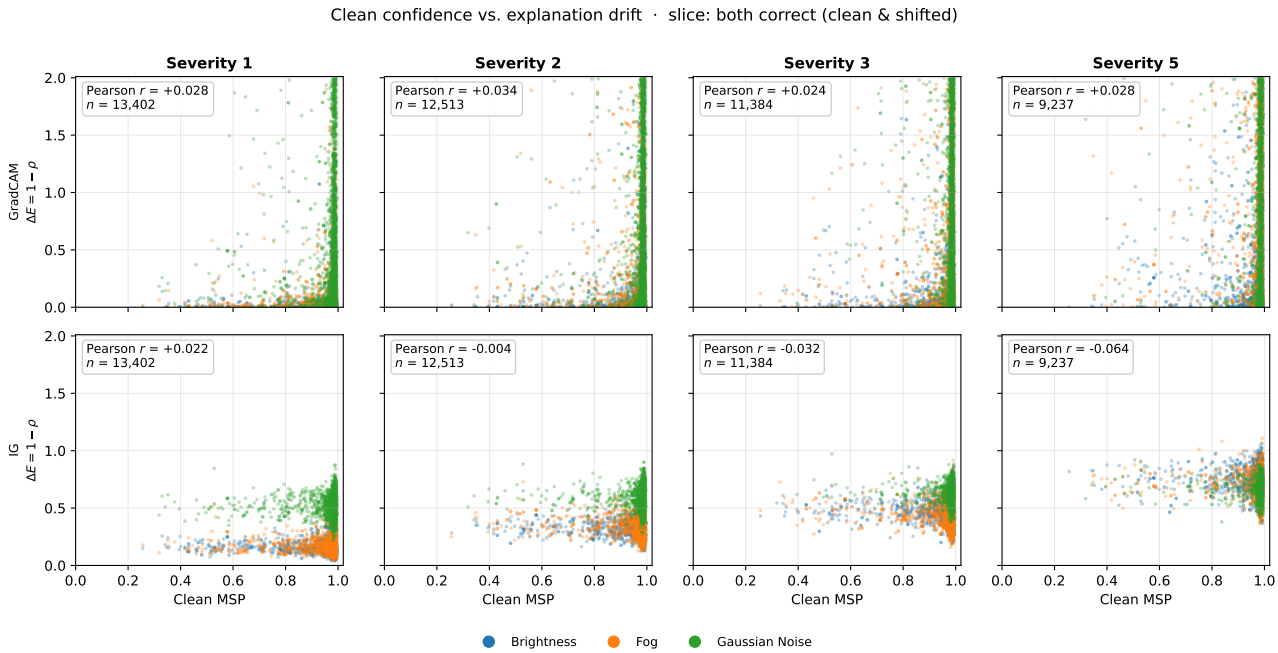

Figure 9: Relationship between clean maximum softmax probability (MSP) and later explanation drift on the both-correct slice. The plot tests whether clean confidence predicts how strongly explanations change under corruption.

At the same time, the point clouds are not fully structureless. Especially for IG, samples cluster by corruption type: at similar confidence levels, Gaussian noise tends to lie above brightness and fog. This suggests that later explanation drift depends more on corruption type than on pre-shift certainty.

Summary. Taken together, these results show that explanation drift and prediction change often increase together as corruption becomes stronger, but they are not interchangeable signals. Substantial drift survives even after prediction-changing samples are removed, this residual drift affects a substantial share of individual samples, and it cannot be predicted from clean-image confidence. The explainer comparison is central to this conclusion: GradCAM shows its clearest slice reduction under Gaussian noise and a more concentrated per-sample drift profile, whereas IG shows the strongest Silent Drift signature under brightness, and the most population-wide drift pattern.

5.3 RQ3: Metric change depends on both the metric and the explainer

To answer RQ3, we compare changes in evaluation scores (ΔQ) against explanation drift (ΔE) across corruption conditions and slices. The decoupling result from Section 5.2 gives this comparison a clear interpretation. If a metric's response to corruption disappears once we restrict the analysis to samples whose predictions remain stable, then the metric is primarily tracking model-behaviour change and is therefore not independent of ΔP. If, by contrast, the response persists on the both-correct slice, where prediction behaviour is held constant by construction, then the metric is reacting to an aspect of the attribution that exists independently of prediction change.

In our setting, only Sparseness yields a stable and interpretable pattern. Sparseness measures how concentrated the attribution mass is, with higher values corresponding to more selective explanations.

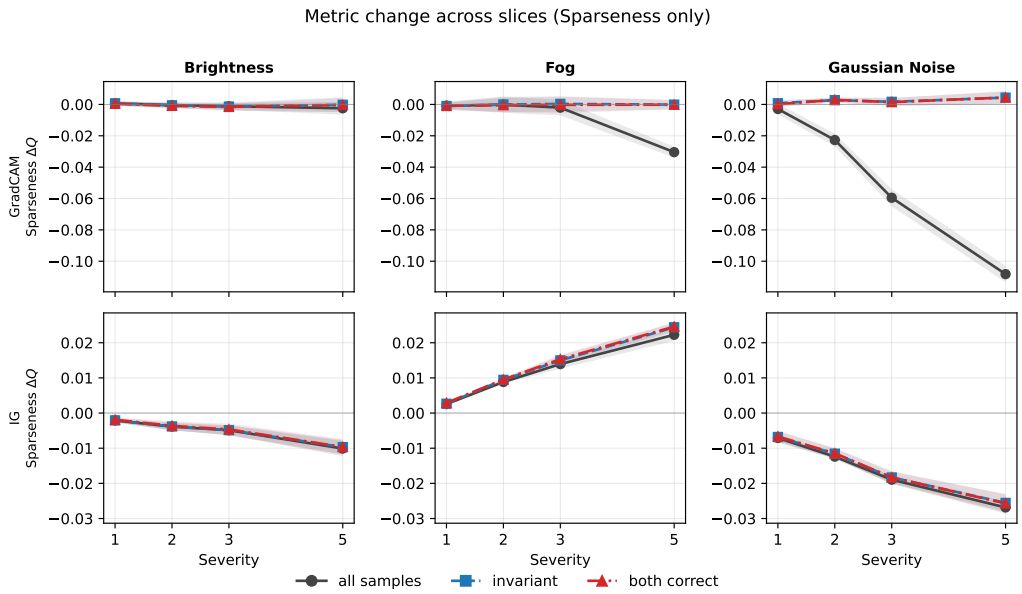

Figure 10: Sparseness change (ΔQ) across the full, invariant, and both-correct slices. This shows whether metric movement persists when prediction change is controlled for.

Figure 10 reports ΔQ for Sparseness across the three slices, for both explainers and all three corruption types. The two explainer rows follow distinct response patterns.

For GradCAM, Sparseness changes substantially under the more disruptive corruptions on the full sample set. The largest excursion is $\Delta Q = -0.108$ under Gaussian noise at severity 5, a 30% relative reduction against the clean baseline of $Q_{\text{clean}} = 0.363$. On the invariant and both-correct subsets, however, this response collapses to within ± 0.005 of zero across every corruption and severity. This suggests that the metric's movement on the full sample set is driven mainly by the subset of samples on which corruption also changes model behaviour.

For IG, the same metric shows the opposite pattern. The three slice trajectories lie almost exactly on top of each other in every IG panel, differing by less than 0.001 in every condition: the metric moves in the same way whether flipped samples are included or excluded. Even on the both-correct slice, ΔQ for Sparseness remains non-zero across corruption types, with the largest relative shift in the underlying score reaching 4%. This pattern is also not consistent with random noise: the trajectories are stable and structured. Under brightness and Gaussian noise, Sparseness systematically *decreases* with drift, so attribution mass becomes more diffuse; under fog, it *increases*, yielding a more concentrated attribution pattern.

The key point is that this movement remains even after prediction change has been removed by construction. For IG, the Sparseness response therefore tracks a reorganisation of the attribution map. The direction of that response depends on the corruption type: IG Sparseness decreases under brightness and Gaussian noise, but increases under fog. This indicates that the metric remains sensitive to corruption-specific structural changes in the attribution map even when predictions stay stable.

Figure 11 reinforces this contrast at the aggregate level. For GradCAM, Sparseness and explanation drift are positively associated across corruption conditions, but the slice analysis shows that this association is concentrated in the subset of samples where corruption also changes prediction behaviour. For IG, the relation is weaker and more heterogeneous, consistent with the fact that Sparseness

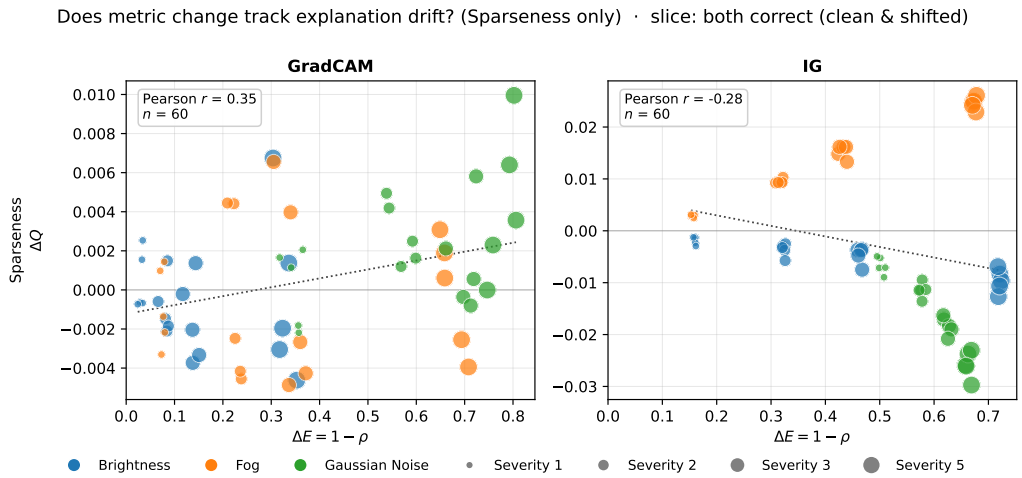

Figure 11: Relation between Sparseness change (ΔQ) and explanation drift (ΔE) on the both-correct slice, testing whether Sparseness continues to move when prediction change is controlled for.

continues to move on controlled subsets while not following a single common direction across all corruptions.

The second metric considered, Faithfulness Correlation, also changes under shift, but its absolute ΔQ values stay below 0.02 on both slices and its behaviour is less structured across slices and corruption conditions. We report it in Appendix E for completeness and draw no conclusions from it.

The RQ3 conclusion is therefore specific rather than general. In the present setup, Sparseness is the only evaluation metric that yields a stable enough signal for interpretation, and even for Sparseness that signal depends strongly on the explainer. For GradCAM, Sparseness mainly reflects model-behaviour change; for IG, it also captures attribution-level change that persists under stable correct predictions. Metric response therefore depends jointly on the evaluation metric and the explainer it is paired with, and cannot be inferred from the metric alone.

5.4 Putting it together: two explainers, two failure profiles

Across the three research questions, GradCAM and IG behave differently enough that they cannot be treated as interchangeable instances of a generic attribution method. Instead, the results point to two distinct response profiles.

For corruption sensitivity (RQ1), the two explainers do not follow a single shared corruption ordering. At severity 5, GradCAM is least affected by brightness and most affected by Gaussian noise. IG shows a different pattern: it already exhibits substantial drift under Gaussian noise at low severity, while the largest severity-driven increases come from the photometric corruptions. The relevant result is therefore not a common ranking of corruption types, but that both the type and the magnitude of explanation drift depend on the explainer.

The same contrast appears in how drift is distributed across samples under stable predictions (RQ2). On the both-correct slice, most GradCAM samples remain close to the clean attribution, while drift is concentrated in a smaller subset whose deviations become increasingly severe as corruption strength rises. IG, by contrast, shifts a much larger share of the sample population by a moderate amount, producing a broader downward movement of the similarity distribution rather than a small set

of extreme cases. Even after prediction-changing samples are removed, drift is therefore not distributed in the same way for the two explainers.

For quality-metric response (RQ3), the difference appears again. GradCAM Sparseness changes on the full sample set, but this response disappears once the analysis is restricted to samples with stable predictions. Its metric movement therefore mainly reflects changes in model behaviour. IG Sparseness behaves differently: it still changes by up to $|\Delta Q| = 0.026$ even on samples whose predictions are stable by construction. Its response therefore captures attribution-level change that is not reducible to model behaviour alone.

Taken together, these results suggest two different failure profiles rather than one shared one. GradCAM fails rarely but severely: most samples remain close to their clean attribution, but the subset that drifts can change dramatically (Section 5.2, Figure 7). For GradCAM, the corruption to which it is least sensitive is also the one associated with the smallest change in prediction behaviour (Section 5.1), and when GradCAM-based evaluation metrics move, they mainly track model-behaviour change (Section 5.3). IG fails broadly but more moderately: drift affects a much larger share of samples (Section 5.2), the method responds strongly to photometric shifts that the model often handles correctly (Section 5.1), and its quality-metric response retains a signal even after model-behaviour change has been controlled for (Section 5.3).

The practical implication is not that one profile is uniformly preferable. Rather, different attribution methods make different failure modes visible. This becomes especially clear in Figure 12. Both panels show the same image, a cat in vegetation, under Gaussian noise at severity 3, and in both cases the clean and corrupted predictions are wrong. What differs is whether the failure is visible in the explanation-drift signal. IG shows a substantial attribution change, with the per-sample Spearman correlation dropping to $\rho = 0.275$. GradCAM, by contrast, remains almost unchanged, with $\rho = 0.998$. The same underlying prediction failure is therefore clearly flagged by one explainer but almost invisible to the other.

Under the present conditions, failure detectability depends on the attribution method. Feature-map-based and pixel-level attribution methods capture different aspects of the model's response to shift and therefore provide different monitoring signals. The results of Sections 5.1 to 5.3 make this dependence explicit: any interpretation based on a single explainer also makes an implicit assumption about which failure modes are captured by that explainer.

6 Conclusion

This report investigated how post-hoc explanations behave under controlled covariate shift and how explanation changes relate to prediction changes and evaluation metrics. Across paired clean-corrupted samples from CIFAR-10-C, analysed with GradCAM and Integrated Gradients on a ResNet-18 classifier, three main findings emerge. First, explanation drift arises systematically under distribution shift for both explainers and increases with corruption severity. The effect is not marginal: drift is already visible at low corruption levels and becomes substantial at higher severities. At the same time, its magnitude and progression depend on the attribution method, indicating that the explainer itself shapes how sensitivity to shift becomes visible. Second, explanation drift is not reducible to predictive failure. Across seven of eight conditions, substantial drift remains when the analysis is restricted to correctly classified samples. The sample-level regime analysis reinforces this result: a large share of samples falls into *Silent Drift*, where predictions remain correct while explanations change substantially. Under the present conditions, explanation stability therefore cannot be inferred from

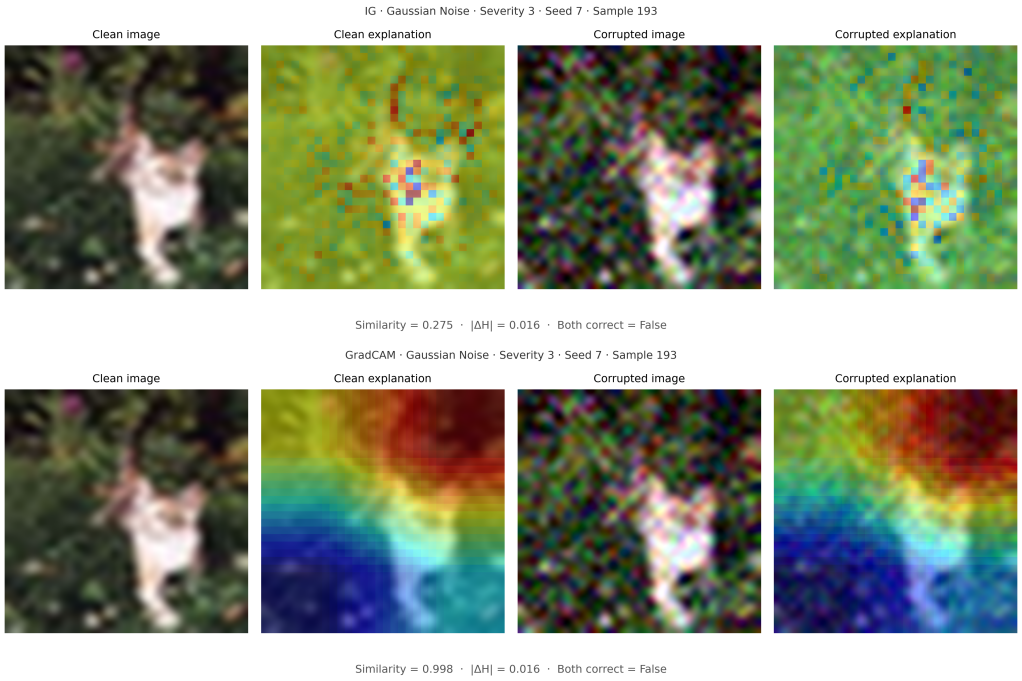

Figure 12: Same failure, different drift visibility. Both rows show the same image under Gaussian noise at severity 3, with wrong predictions before and after corruption. IG shows substantial attribution change ($\rho = 0.275$), whereas GradCAM remains almost unchanged ($\rho = 0.998$).

prediction stability alone. Clean confidence is not a reliable predictor of later drift either. Third, metric response under shift depends jointly on the metric and the explainer. In the present setup, Sparseness is the only metric that yields a comparatively stable and interpretable pattern, but even this effect is explainer-specific. For GradCAM, Sparseness largely collapses once prediction changes are controlled for, whereas for IG it continues to move on the both-correct slice, indicating sensitivity to attribution-level reorganisation rather than only to model-behaviour change. Metric movement under shift therefore cannot be treated as a generic proxy for either explanation drift or model failure. Taken together, these findings support the central claim of this report: performance shift, explanation shift, and metric shift are related, but not interchangeable. Their relation depends on the explainer, the corruption, and the evaluation metric. Practically, this means that explanation-based monitoring should not rely on prediction stability alone, and conclusions drawn from a single attribution method should be interpreted with explicit awareness of explainer-specific behaviour. More specifically, GradCAM and IG expose different failure profiles: GradCAM is often stable for many samples but can shift abruptly in a smaller subset, whereas Integrated Gradients shows broader drift across the sample population.

Limitations and future work. These conclusions are bounded by the present setup: a single model architecture and dataset, four corruption types at discrete severity levels, two attribution methods, and a metric analysis that ultimately rests on a single comparatively stable metric. Future work should therefore extend the analysis to additional architectures, shift types, and evaluation criteria. Two directions are particularly important: evaluating drift-based monitoring across multiple attribution methods in parallel, and validating explanation-quality metrics on controlled subsets so that responses

driven by model-behaviour change can be distinguished from responses driven by attribution-level change.

AI Use Disclosure

During the preparation of this paper, ChatGPT (OpenAI) and Claude (Anthropic) were used to assist with code debugging, code review, proofreading, and language polishing. All suggestions were reviewed, verified, and revised as needed by the author, who takes full responsibility for the final submitted version.

Code Availability

The source code used for data processing, model evaluation, and analysis is available at `https://github.com/MicheleMarschner/explanation_shift.git`.

Appendix A. Additional experimental design details

A.1 Overview of computed measures

This appendix summarizes the main measures computed in the empirical analysis, their interpretation, and their level of aggregation. Table 1 provides a compact reference for the measures introduced in Sections 4.2 and 4.4. At the aggregate level, these contribute to the three central change variables used throughout the paper: ΔP for predictive change, ΔE for explanation drift, and ΔQ for explanation-quality metric change. Reporting decisions about which measures are retained in the main text are discussed separately in Section 4.5 and Appendix B.

A.2 Metric definitions and evaluation tests

In the experiment, XAI evaluation is operationalised along four dimensions: complexity, robustness, faithfulness, and randomisation sensitivity.

Complexity: Sparseness. For complexity, we use Sparseness, which captures whether attribution mass is concentrated on a small subset of input features or distributed more diffusely across the input. It thus operationalises complexity in terms of how compact and selective, rather than broad and noisy, an explanation is. Higher values indicate more concentrated attributions, whereas lower values indicate more diffuse attribution patterns. Since Sparseness does not rely on additional input perturbations, it is not perturbation-sensitive.

Robustness: AvgSensitivity and Local Lipschitz Estimate. Robustness is assessed with AvgSensitivity or, alternatively, a Local Lipschitz Estimate. AvgSensitivity measures how much an explanation changes on average under repeated small perturbations of the input, whereas the Lipschitz-based variant targets the strongest local change that can occur in response to a small input variation. Both tests therefore probe whether minor, non-substantive input changes leave the explanation approximately stable. For both metrics, lower values indicate greater explanation stability. Because both metrics are explicitly defined through repeated perturbations of the input, they are highly perturbation-sensitive by construction.

Faithfulness: FaithfulnessCorrelation. To evaluate faithfulness, we use FaithfulnessCorrelation. This metric perturbs subsets of features, measures the resulting change in model output, and relates these output changes to the relevance scores assigned by the explanation. In this way, it tests whether features deemed important by the explainer are also those most strongly associated with the model's behaviour. Higher values indicate that attribution scores align more closely with the observed effect of perturbing features. FaithfulnessCorrelation is perturbation-sensitive, since its score depends directly on how feature perturbations are implemented.

Randomisation: RandomLogit. Finally, randomisation sensitivity is measured with RandomLogit. Rather than perturbing the input, this test compares the explanation for the actual prediction target with a reference explanation for a randomly chosen non-target class. It thereby examines whether the attribution is specifically tied to the decision being explained, rather than remaining largely unchanged when the explanation target is randomised. In Quantus, lower values are preferable, since a smaller distance indicates that the explanation is less affected by target randomisation and thus more specifically tied to the original target class. Since RandomLogit does not perturb the input itself, it is less directly perturbation-sensitive; instead, its sensitivity lies in the randomisation of the explanation target.

Measure	Description	Scope
Shift strength		
MMD^2	Distributional distance between clean and corrupted representations, used as a continuous indicator of shift magnitude.	corruption × severity
Prediction level		
Accuracy	Predictive performance on clean and corrupted data; used to derive aggregate ΔP.	aggregate
Prediction entropy	Entropy of the predictive distribution, used as an uncertainty indicator.	pairwise, aggregated
ΔH	Change in prediction entropy between clean and corrupted inputs.	pairwise, aggregated
Probability shift	Change in predicted-class probability between clean and corrupted inputs.	pairwise, aggregated
Margin shift	Change in prediction margin between clean and corrupted inputs.	pairwise, aggregated
Explanation level		
Spearman rank correlation	Rank-based similarity between paired attribution maps; captures agreement in relative feature ordering.	pairwise
Cosine similarity	Directional similarity between paired attribution vectors.	pairwise
Top-k IoU	Overlap among the most highly attributed features in paired attribution maps.	pairwise
Evaluation level		
Sparseness	Explanation-quality score computed with Quantus; measures how concentrated the attribution mass is.	aggregate by condition
Faithfulness Correlation	Explanation-quality score computed with Quantus; measures the association between attributed importance and output change under perturbation.	aggregate by condition
Random Logit	Explanation-quality score computed with Quantus; tests whether explanations change when model outputs are randomly reassigned.	aggregate by condition
Avg.Sensitivity, Local Lipschitz Estimate	Explanation-quality scores computed with Quantus to assess the stability of explanations under perturbations.	aggregate by condition
MetaQuantus metrics	Supplementary diagnostics based on MetaQuantus, used to assess aspects of metric reliability across conditions.	aggregate by condition

Table 1: Overview of the main measures computed in the empirical analysis, grouped by analytical level.

Appendix B. Practical evaluation metric limitations

A broader set of Quantus and MetaQuantus metrics was explored during development, but some proved numerically unstable or computationally infeasible under the full experimental setup and were therefore excluded from the reported analysis. Table 2 summarizes the affected metrics and the resulting reporting decisions.

Metric	Issue	Consequence
Average Sensitivity	Frequent NaN values in Quantus runs	Excluded from the final analysis
Random Logit	Frequent NaN values and unstable outputs	Excluded from the final analysis
Local Lipschitz Estimate	Computational cost too high for the full experimental grid	Not used systematically
Faithfulness Correlation	Numerically retained, but less stable and less structured across conditions	Reported only in the appendix
Sparseness	Stable and computationally feasible	Used as the main reported evaluation metric
Meta-evaluation with MetaQuantus	Only feasible with substantially fewer iterations; outputs remained unstable and largely uninformative	Excluded from the final analysis

Table 2: Practical limitations affecting metric selection for the metric-level evaluation.

These limitations primarily affect the scope of the ΔQ analysis, which should therefore be interpreted as applying only to the subset of metrics that remained stable and feasible under the full experimental grid.

Additional results

The following appendices collect additional empirical results that support the main text findings in Section 5. They cover metric diagnostics motivating the primary drift measures (Appendix C), robustness checks for the decoupling findings (Appendix D), excluded and null-result evaluation metrics (Appendix E), and full numerical tables for the reported figures (Appendix F).

Appendix C. Metric redundancy and metric choices

Section 5 adopts $\Delta E = 1 - \rho$ and $|\Delta H|$ as the primary drift measures on the basis of empirical redundancy between the candidate metrics. This appendix documents that redundancy at two levels: first, through the correlation structure between the candidate measures; and second, through the stability of the RQ1 drift trajectories under each alternative similarity measure.

Pairwise correlations between candidate drift measures

Figure 13 reports pairwise Spearman correlations between six drift measures: three explanation-drift measures ($1-\cos$, $1-\rho$, $1-\text{IoU}$) and three confidence-drift measures ($|\Delta H|$, $|\Delta p|$, $|\Delta\text{margin}|$), pooled across all samples (left) and broken down by corruption type (right).

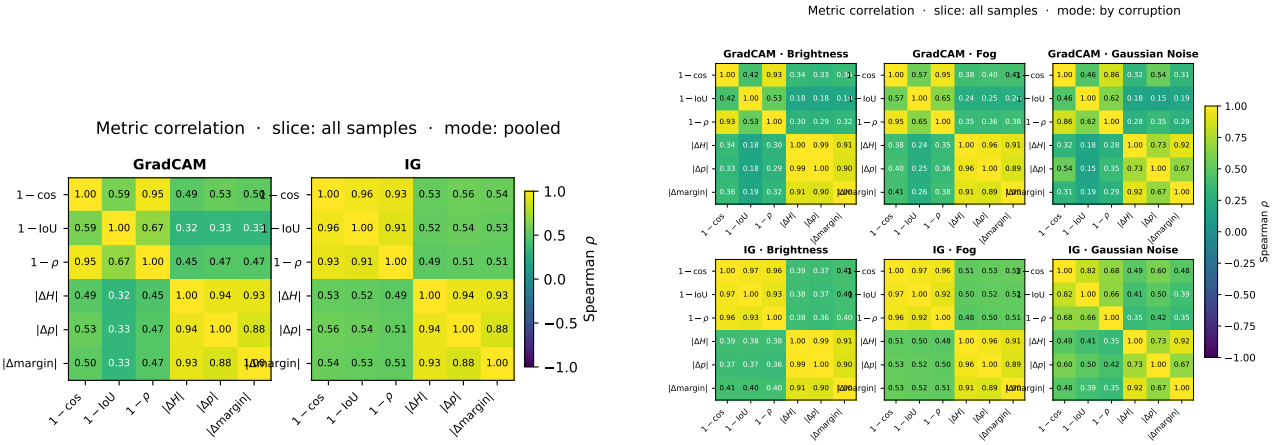

Figure 13: Pairwise Spearman correlations between drift metrics. Left: pooled across all samples. Right: broken down by corruption type.

Confidence-drift measures are fully redundant. The three confidence measures correlate at 0.89-0.99 across every corruption, slice, and explainer. A single uncertainty measure is therefore sufficient for the main analysis, and we use $|\Delta H|$ because it is the most interpretable of the three.

Explanation-drift measures are redundant for IG but not fully for GradCAM. The three similarity-based drift measures are also strongly correlated overall. For IG, the pooled pairwise correlations are 0.92-0.97 across corruptions. For GradCAM, $1 - \cos$ and $1 - \rho$ remain closely aligned, whereas $1 - \text{IoU}$ is less tightly coupled to both (0.42-0.57 against $1 - \cos$, 0.53-0.67 against $1 - \rho$). This separation is stable across brightness, fog, and Gaussian noise and is therefore not a corruption-specific artefact.

We interpret this difference structurally. GradCAM maps are low-resolution and spatially diffuse, so top-k overlap reacts more strongly to localisation shifts that continuous similarity measures smooth over the full map. IG maps are higher-resolution and sparse enough that all three measures remain closely aligned. The corruption-wise breakdown supports the same interpretation and shows most clearly under Gaussian noise that GradCAM $1 - \text{IoU}$ behaves as a distinct but still related drift measure. We therefore report $1 - \text{IoU}$ trajectories as a complementary robustness check for GradCAM in the appendix.

Drift trajectories under alternative similarity measures

The correlation analysis above suggests that the three drift measures are largely interchangeable. Figure 14 complements this result by showing that the qualitative RQ1 conclusions are preserved across all three measures.

Across all three measures, the main pattern is unchanged. For both explainers, drift increases monotonically with severity, and the corruption ordering from Section 5.1 is preserved. For GradCAM, brightness remains the mildest corruption and Gaussian noise the most disruptive under $1 - \rho$, $1 - \cos$, and $1 - \text{IoU}$; for IG, the three corruptions again converge at high severity under all three measures.

What changes is mainly the visual spread of the curves. For GradCAM, $1 - \cos$ compresses the dynamic range relative to $1 - \rho$: GradCAM Gaussian noise reaches only $\Delta E \approx 0.50$ at severity 5, compared with 0.86 under Spearman. For IG, $1 - \text{IoU}$ makes the high-severity convergence even

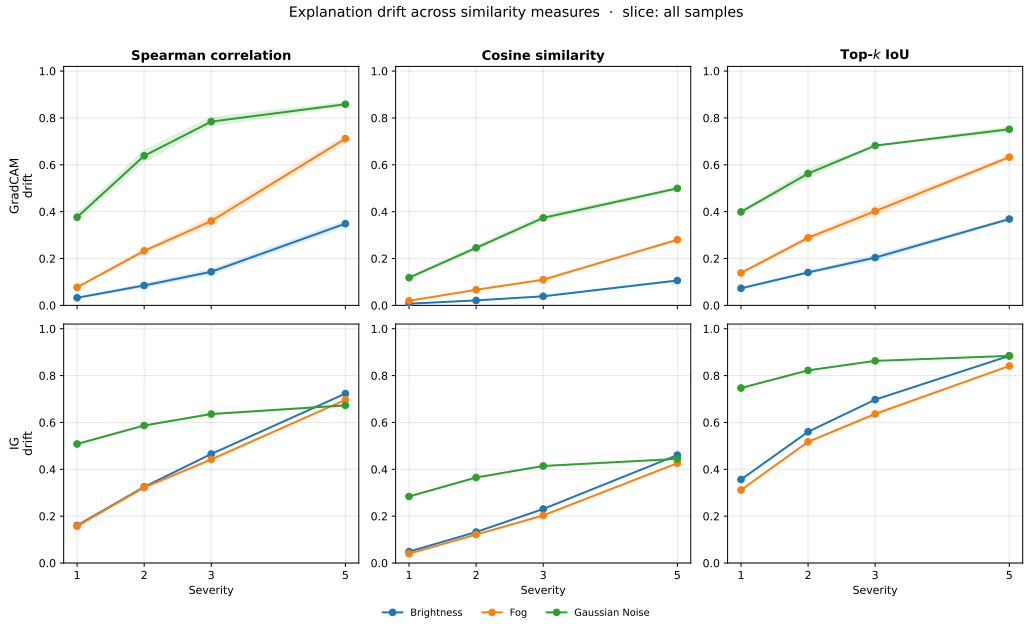

Figure 14: Explanation drift under three similarity measures. Columns: $1-\rho$ (left), $1-\cos$ (middle), $1-\text{IoU}$ (right). Rows: GradCAM (top), IG (bottom). All trajectories computed on the full sample set across five seeds.

tighter: IG brightness and IG Gaussian noise both reach $\Delta E \approx 0.88$ at severity 5. These differences do not change the interpretation of the results and are consistent with the correlation findings above.

Appendix D. Robustness of the decoupling findings

Section 5.2 presents the decoupling claim on the both-correct slice with the drift metric $\Delta E = 1 - \rho$ and a fixed threshold for the prediction-drift regimes of $q_{0.75} = 0.632$. This appendix shows that the qualitative conclusions are robust to the choice of slice, threshold, and similarity metric.

Invariant vs. both-correct slice

The main text reports figures on the both-correct slice rather than the larger invariant slice because the two are numerically almost identical. Across all 24 explainer-corruption-severity conditions, the absolute difference $|\Delta E_{\text{invariant}} - \Delta E_{\text{both-correct}}|$ has a maximum of 0.006 (GradCAM fog, severity 5) and a mean of 0.001, well within the variation across seeds reported in the main text. The two slices differ only in samples that were already misclassified on the clean image, and that subset is small enough that removing it has no measurable effect on drift. The main text therefore reports only the both-correct slice throughout, since using the invariant slice would not change the conclusions.

Threshold sensitivity of the prediction-drift regimes

Recomputing the prediction-drift regime analysis with alternative drift thresholds at $q_{0.50}$ and $q_{0.90}$ leaves the qualitative interpretation unchanged: Silent Drift remains the dominant regime for IG under photometric corruptions at high severity, and the relative ordering of the main configurations

is preserved. The result is therefore not tied to a single arbitrary cutoff, but reflects a stable property of the underlying per-sample drift distributions.

Prediction-drift regimes under $\Delta E = 1 - \text{IoU}$

Figure 15 reports the prediction-drift regime analysis under the alternative drift metric $\Delta E = 1 - \text{IoU}$, with pooled drift threshold $q_{0.75} = 0.903$. Exact shares at severity 5 are reported in Table 4 (panel b) alongside the values for $\Delta E = 1 - \rho$ for direct comparison.

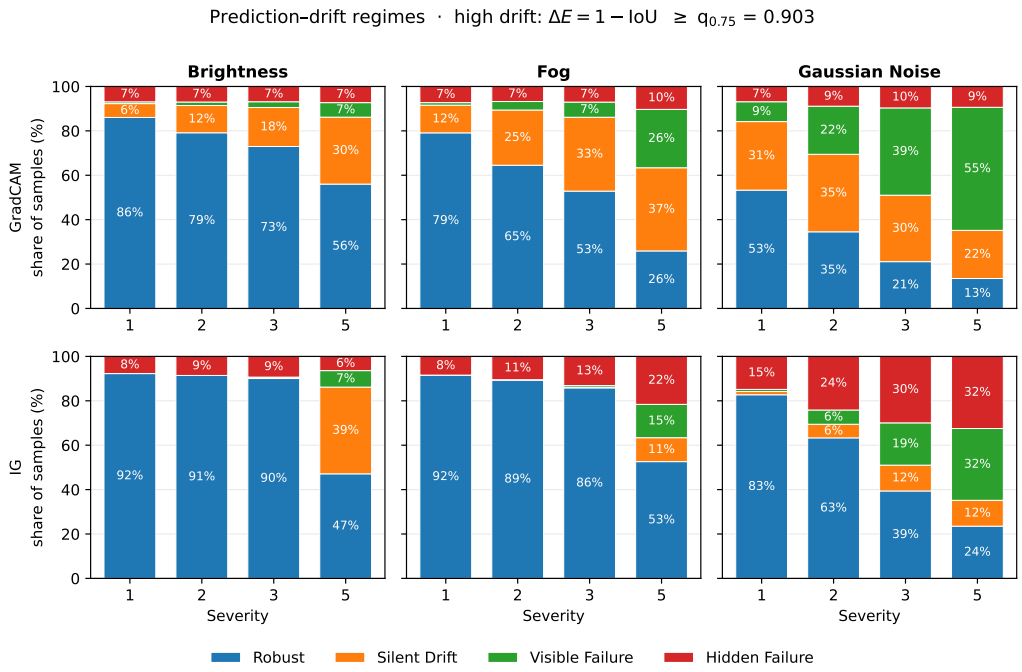

Figure 15: Prediction-drift regime shares across severities under the alternative drift metric $\Delta E = 1 - \text{IoU}$ with pooled threshold $q_{0.75} = 0.903$.

The overall picture is the same under the alternative similarity metric. The main result from Section 5.2 still holds: explanation drift can remain substantial even when predictions stay unchanged. At severity 5, Silent Drift is still present in every explainercorruption configuration, with shares between 10.8% and 39.1%. Severity also shifts regime mass in the same overall direction as under $1 - \rho$, and Gaussian noise still moves mass into the failure regimes for both explainers. What changes is not the existence of the effect, but its magnitude in specific configurations.

The strongest metric-dependent changes appear under photometric corruptions. For IG, Silent Drift is smaller under $1 - \text{IoU}$ than under $1 - \rho$, dropping from 70.6% to 39.1% under brightness and from 41.5% to 10.8% under fog; in both cases, Robust becomes the largest regime. For GradCAM, the pattern reverses: under IoU, Silent Drift increases for both brightness (15.6% $\rightarrow$ 30.2%) and fog (27.5% $\rightarrow$ 37.5%). The qualitative decoupling result therefore remains the same, but the exact regime shares depend on the metric.

These differences come from how the two metrics capture drift. For IG under photometric corruptions, many samples cross the high-drift threshold under $1 - \rho$, but fewer do so under IoU because the same attribution maps appear sparser there. For GradCAM, the pattern is reversed: continuous

similarity measures react less strongly to localisation shifts, while top-k overlap is more sensitive to them. As a result, more GradCAM samples are classified as high drift under IoU than under $1 - \rho$. This is consistent with the structural metric analysis in Appendix C.

A second difference appears in the severity trajectory. For IG under photometric corruptions, IoU concentrates the transition out of the Robust regime almost entirely between severity 3 and severity 5. Under brightness, the Robust share remains high through severity 3 (90%) and then drops sharply to 47% at severity 5. Under fog, the same late drop appears, from 86% to 53%. Under $1 - \rho$, the same configurations decline more gradually across all severities.

The main conclusions therefore remain unchanged: (i) Silent Drift exists at non-trivial levels in every explainercorruption configuration at severity 5; (ii) severity monotonicity holds under both metrics for both explainers; and (iii) Gaussian noise shifts mass into the failure regimes for both explainers under both metrics. What is metric-specific is the exact size of Silent Drift in some configurations, especially for IG under photometric corruptions, and the severity at which it becomes dominant.

Clean confidence check on the full sample

Figure 16 complements Figure 9 by extending the clean-confidence vs. drift analysis from the both-correct slice to the full sample set.

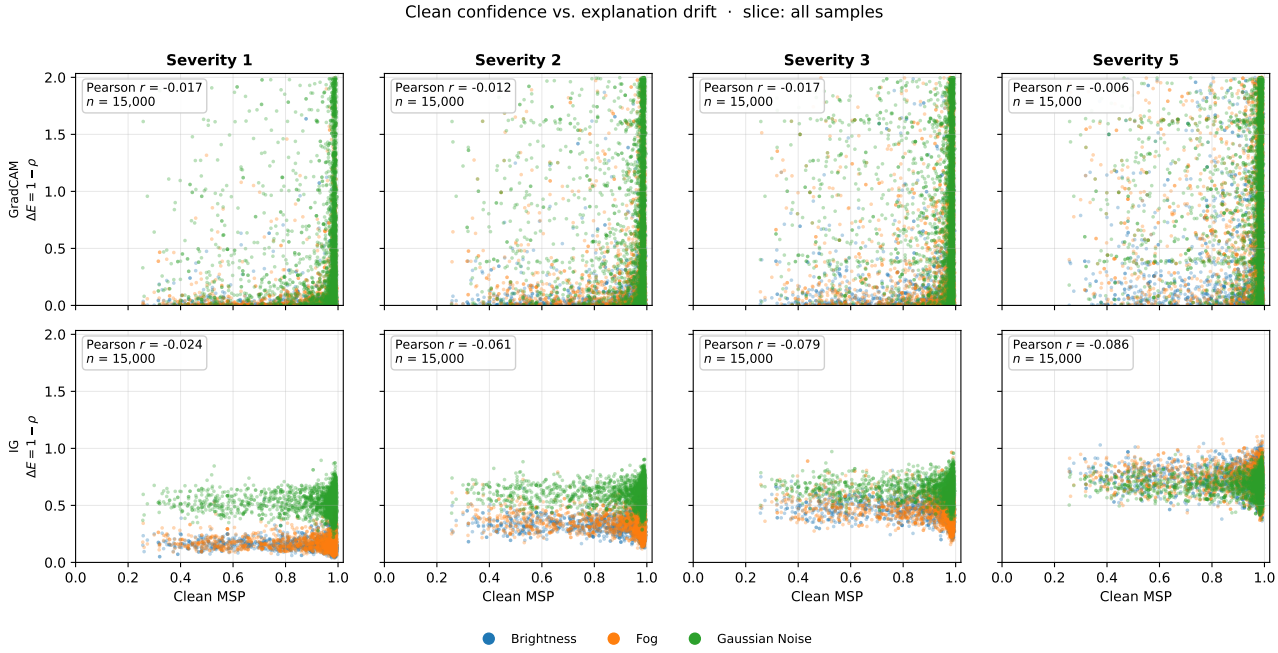

Figure 16: Clean MSP vs. per-sample ΔE on the full sample set.

Across all four severities and both explainers, Pearson correlations remain close to zero ($[-0.086, -0.006]$). Clean confidence therefore does not predict per-sample drift on either the both-correct slice or the full sample set.

Appendix E. Excluded and null-result quality metrics

Section 5.3 reports only the Sparseness metric, which was stable enough to support interpretation in the main text. Two further metrics were initially considered but are not used there: Faithfulness Correlation produced no interpretable signal, as discussed below, and Model Parameter Randomisation did not converge under our configuration.

Faithfulness Correlation

Figure 17 reports Faithfulness Correlation ΔQ across the three slices used in the main text (upper rows of the full figure), for both explainers and all three corruption types.

Faithfulness Correlation does change under shift, but its absolute ΔQ values remain small ($[-0.02, +0.05]$) across all configurations, and its behaviour is much less structured than for Sparseness. For IG under Gaussian noise, for example, an apparent peak at severities 1 and 2 collapses again at severities 3 and 5, producing a non-monotonic pattern that Sparseness does not show. Under the present configuration, it is therefore unclear whether the metric is simply too noisy or whether it tracks something largely independent of explanation drift. We report it only as supplementary evidence and do not use it to support the main RQ3 conclusions.

ΔQ vs. ΔE correlation

Figure 18 plots condition-level ΔQ against ΔE for both metrics on the both-correct slice.

The four Pearson correlations reinforce the difference between the two metrics, but only one corresponds to a practically meaningful signal. For GradCAM Sparseness, $r = +0.35$, yet all $|\Delta Q|$ values remain below 0.005 across conditions, so the association is statistically detectable but practically negligible. For IG Sparseness, $r = -0.28$ occurs on substantially larger $|\Delta Q|$ values up to 0.026, making it the only setting where the condition-level correlation aligns with a practically meaningful shift.

For Faithfulness Correlation, both explainers show positive r (0.28 for GradCAM, 0.44 for IG), but the underlying ΔQ values remain small and unstable, and their variation across seeds includes zero in every condition. These correlations therefore do not support a meaningful interpretation.

Appendix F. Full numerical results

This appendix collects the numerical results underlying the figures in Sections 5.1 through 5.3.

Prediction-drift regime shares. Table 4 reports the prediction-drift regime shares at severity 5 under both drift metrics: the $1 - \rho$ threshold $q_{0.75} = 0.632$ used in the main text (upper panel), and the $1 - \text{IoU}$ threshold $q_{0.75} = 0.903$ used in the robustness check of Appendix D (lower panel). The two metrics define high drift relative to distributions with different tail behaviour, which is why the absolute shares differ between panels; Appendix D discusses the implications.

Seed variability. Across all tabulated conditions, the maximum seed standard deviation is 0.028 on ΔE (GradCAM, Gaussian noise, severity 2, all slice) and 0.006 on Sparseness ΔQ. In most conditions, the ΔE seed standard deviation stays below 0.015. We therefore do not report separate analyses of variation across seeds beyond the ranges shown in the main-text figures.

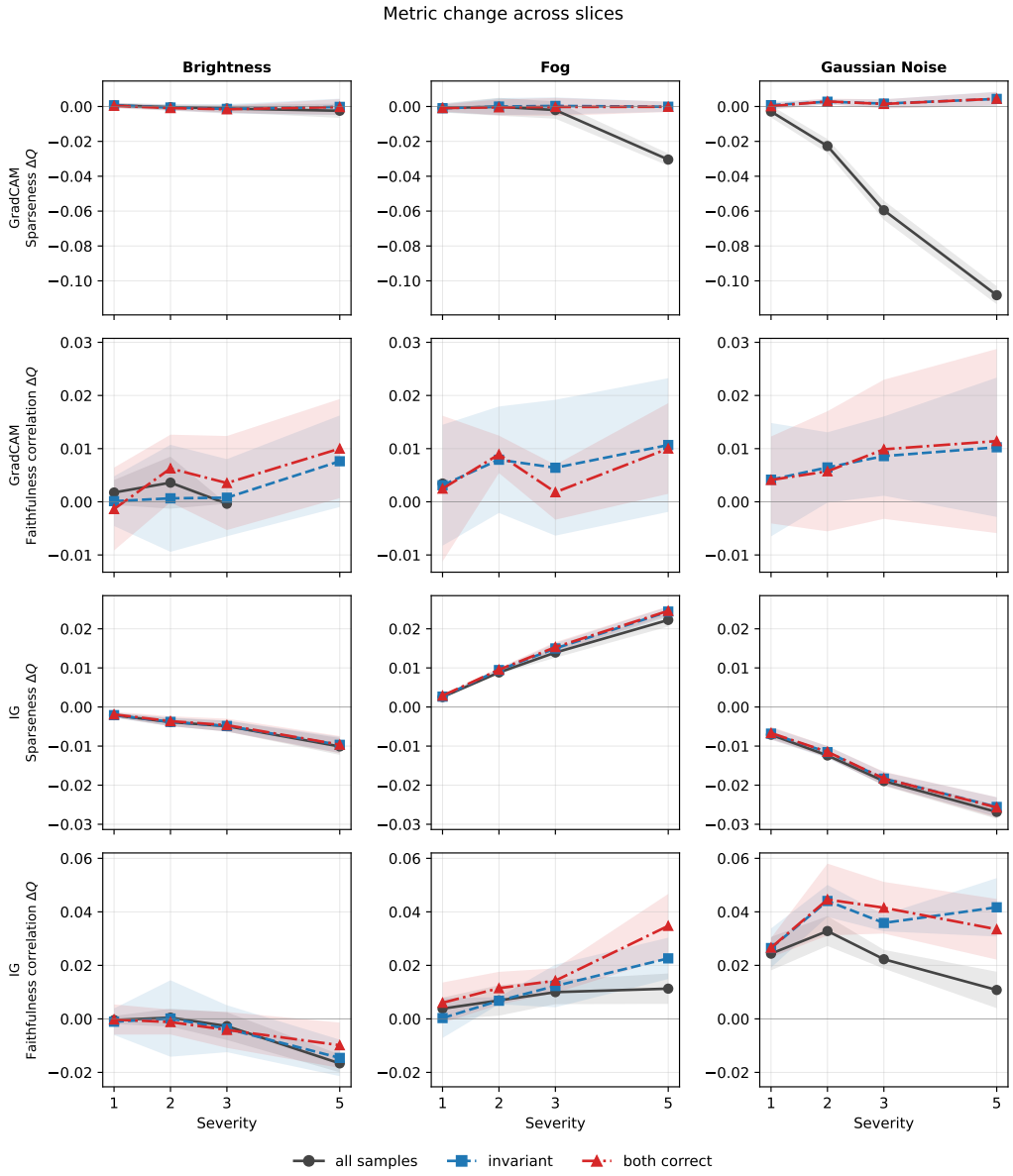

Figure 17: Metric change ΔQ across slices for both explainers and both metrics. Sparseness rows (first and third) are analysed in Section 5.3.

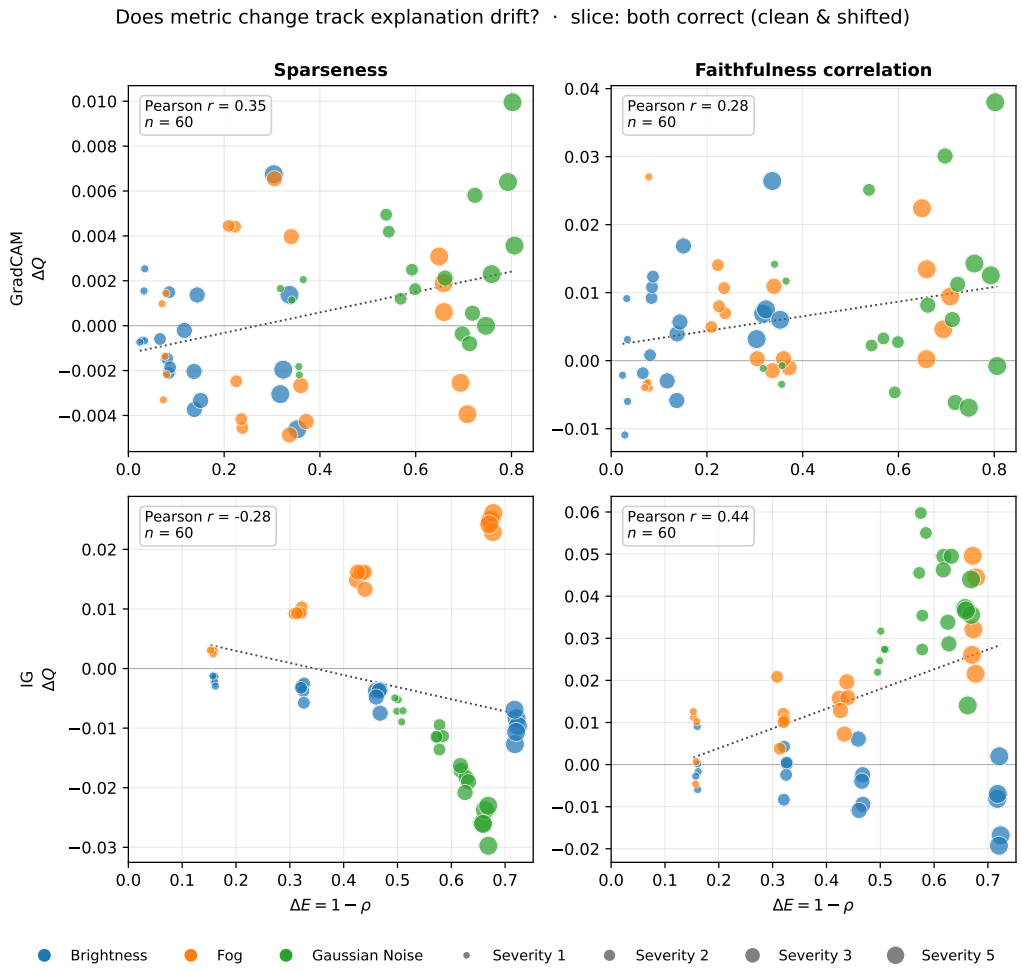

Figure 18: Condition-level ΔQ vs. ΔE on the both-correct slice. Each marker is one seed of one (corruption, severity) condition.

Explainer	Corruption	Severity	ΔE_{all}	ΔE_{inv}	ΔE_{bc}
GradCAM	Brightness	1	0.032 ± 0.005	0.032 ± 0.004	0.031 ± 0.004
GradCAM	Brightness	2	0.085 ± 0.011	0.083 ± 0.009	0.081 ± 0.009
GradCAM	Brightness	3	0.143 ± 0.013	0.141 ± 0.012	0.137 ± 0.013
GradCAM	Brightness	5	0.349 ± 0.016	0.330 ± 0.018	0.327 ± 0.019
GradCAM	Fog	1	0.077 ± 0.005	0.076 ± 0.004	0.076 ± 0.004
GradCAM	Fog	2	0.233 ± 0.010	0.227 ± 0.011	0.226 ± 0.012
GradCAM	Fog	3	0.360 ± 0.021	0.345 ± 0.025	0.343 ± 0.025
GradCAM	Fog	5	0.712 ± 0.019	0.668 ± 0.024	0.674 ± 0.026
GradCAM	Gaussian noise	1	0.376 ± 0.019	0.348 ± 0.018	0.347 ± 0.019
GradCAM	Gaussian noise	2	0.638 ± 0.026	0.569 ± 0.028	0.568 ± 0.027
GradCAM	Gaussian noise	3	0.784 ± 0.021	0.703 ± 0.023	0.703 ± 0.025
GradCAM	Gaussian noise	5	0.858 ± 0.011	0.781 ± 0.023	0.781 ± 0.027
IG	Brightness	1	0.160 ± 0.002	0.160 ± 0.002	0.160 ± 0.002
IG	Brightness	2	0.325 ± 0.003	0.324 ± 0.003	0.324 ± 0.003
IG	Brightness	3	0.465 ± 0.004	0.465 ± 0.004	0.464 ± 0.004
IG	Brightness	5	0.723 ± 0.002	0.721 ± 0.003	0.720 ± 0.002
IG	Fog	1	0.157 ± 0.003	0.157 ± 0.003	0.156 ± 0.003
IG	Fog	2	0.322 ± 0.006	0.319 ± 0.006	0.317 ± 0.006
IG	Fog	3	0.442 ± 0.008	0.434 ± 0.008	0.432 ± 0.007
IG	Fog	5	0.696 ± 0.003	0.677 ± 0.003	0.674 ± 0.003
IG	Gaussian noise	1	0.508 ± 0.006	0.503 ± 0.006	0.503 ± 0.006
IG	Gaussian noise	2	0.587 ± 0.005	0.578 ± 0.005	0.578 ± 0.005
IG	Gaussian noise	3	0.636 ± 0.004	0.625 ± 0.006	0.624 ± 0.006
IG	Gaussian noise	5	0.673 ± 0.004	0.664 ± 0.005	0.663 ± 0.005

Table 3: Explanation drift by condition, reported as mean ± seed SD across five seeds. Columns correspond to the three sample slices used in the main text: full sample set (all), invariant, and both-correct (bc). Referenced from Figures 3 and 4.

Explainer	Corruption	Robust (%)	Silent Drift (%)	Visible Fail. (%)	Hidden Fail. (%)
(a) $\Delta E = 1 - \rho$*, threshold* $q_{0.75} = 0.632$					
GradCAM	Brightness	70.6	15.6	4.1	9.7
GradCAM	Fog	35.9	27.5	18.9	17.7
GradCAM	Gaussian noise	17.5	17.7	39.6	25.2
IG	Brightness	15.6	70.6	12.0	1.8
IG	Fog	21.9	41.5	30.7	5.9
IG	Gaussian noise	12.1	23.1	46.1	18.7
(b) $\Delta E = 1 - IoU$*, threshold* $q_{0.75} = 0.903$					
GradCAM	Brightness	56.0	30.2	6.5	7.3
GradCAM	Fog	25.9	37.5	26.3	10.3
GradCAM	Gaussian noise	13.5	21.7	55.4	9.4
IG	Brightness	47.1	39.1	7.4	6.4
IG	Fog	52.6	10.8	15.0	21.6
IG	Gaussian noise	23.5	11.7	32.3	32.5

Table 4: Prediction-drift regime shares at severity 5 under both drift metrics. Panel (a): $1 - \rho$, used in Section 5.2 and Figure 6. Panel (b): $1 - \text{IoU}$, used in Figure 15 of Appendix D.

Explainer	Corruption	Sev	Q_{clean}	Q_{shifted}	ΔQ_{all}	ΔQ_{bc}	ΔE_{bc}
GradCAM	Brightness	1	0.363	0.364	+0.001	+0.000	0.031
GradCAM	Brightness	5	0.363	0.361	−0.002	−0.000	0.327
GradCAM	Fog	1	0.363	0.362	−0.001	−0.001	0.076
GradCAM	Fog	5	0.363	0.333	−0.030	−0.000	0.674
GradCAM	Gaussian noise	1	0.363	0.360	−0.003	+0.000	0.347
GradCAM	Gaussian noise	5	0.363	0.255	−0.108	+0.004	0.781
IG	Brightness	1	0.604	0.601	−0.002	−0.002	0.160
IG	Brightness	5	0.604	0.593	−0.010	−0.010	0.720
IG	Fog	1	0.604	0.606	+0.003	+0.003	0.156
IG	Fog	5	0.604	0.626	+0.022	+0.025	0.674
IG	Gaussian noise	1	0.604	0.596	−0.007	−0.007	0.503
IG	Gaussian noise	5	0.604	0.577	−0.027	−0.026	0.663

Table 5: Sparseness ΔQ by condition at severities 1 and 5, reported separately on the full sample set (all) and the both-correct slice (bc). The clean baseline Q_{clean} is identical across all conditions for each explainer by construction. Reference ΔE on the both-correct slice is included for context. Referenced from Figure 10.

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